

Digital Enforcement and Tax Frictions
Evidence from the VAT Registration Threshold in Mongolia

Presented by

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Abstract

This study examines the effect of Mongolia's 2016 ebarimt program, an electronic receipt system designed to strengthen VAT enforcement in business-to-consumer (B2C) transactions. Introduced alongside an increase in the VAT registration threshold from MNT 10 million to MNT 50 million, the program rewarded consumers for registering purchases and created an independent audit trail for retail sales. To identify its effect, I combine a polynomial counterfactual bunching estimator with a difference-in-differences design using binned corporate income tax data covering 2014 to 2023. Threshold bunching declined substantially more in consumer-facing sectors than in business-to-business (B2B) sectors after the reform. The decline accumulated over several years rather than appearing at once, consistent with the gradual uptake of a new monitoring technology. The evidence indicates that consumer-side digital auditing can narrow the B2C enforcement gap that has long constrained VAT collection in low-capacity tax administrations.

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Chapter 1

Introduction

Value-added tax is the largest source of tax revenue for developing countries, but compliance at the final stage of sale is hard to secure. The tax is self-enforcing in business-to-business (B2B) transactions, where both seller and buyer report the sale, but the chain breaks in business-to-consumer (B2C) transactions because consumers have no incentive to report their purchases. B2C sellers therefore face a lower risk of detection and a stronger incentive to underreport revenue to stay below the VAT registration threshold. The gap is most acute in transitional economies, where tax administrations lack the resources for costly audits. Mongolia before 2016 was a case in point: an economy dependent on mining and exposed to commodity-price volatility, with a persistent informal sector and no independent way to verify retail sales.

Two VAT reforms took effect together in January 2016: the registration threshold rose from 10 million (about 4,700 USD)¹ to 50 million MNT (about 23,000 USD), and the government launched ebarimt, an electronic receipt system that rewards consumers for registering purchases and so creates an audit trail for B2C transactions (Chapter 3.1). Because the reforms were simultaneous, identifying the effect of ebarimt requires variation in its relevance across sectors. Retail and food services sell mainly to final consumers and are directly exposed; wholesale and manufacturing transact mainly with other firms, where bilateral invoicing already generates verifiable records, and so serve as a comparison group. If ebarimt strengthened compliance, bunching at the VAT threshold should fall more in B2C than in B2B sectors after the reform.

To test this, I combine a polynomial counterfactual bunching estimator with a difference-in-differences design that exploits cross-sector variation in consumer exposure, using binned administrative corporate income tax returns for 2014 to 2023. I find that bunching fell substantially more in B2C than in B2B sectors after the reform, and the result holds across alternative samples and under small-sample cluster inference. The gap did not close at the reform date but widened over time, with the differential largest in 2018

1. Unless otherwise noted, USD equivalents are calculated using the average annual MNT/USD exchange rate for the relevant year based on data from the Bank of Mongolia (Bank of Mongolia, n.d.).

and again in 2023, a path more consistent with the gradual diffusion of ebarimt than with an immediate enforcement shock.

The study contributes to two literatures. It provides evidence that consumer-side monitoring can narrow the B2C enforcement gap, identified within a threshold-bunching framework rather than from reported-liability data alone. And it extends the VAT bunching literature to a transitional economy, documenting a pre-reform B2C–B2B compliance gap that narrows as digital receipt monitoring takes hold.

The remainder of the thesis is organized as follows. Chapter 2 reviews the literature on VAT threshold bunching and consumer-side enforcement. Chapter 3 sets up the empirical analysis, Chapter 4 presents the results, and Chapter 5 discusses their implications and concludes.

Chapter 2

Literature review

Tax thresholds give firms an incentive to stay below the cutoff to avoid additional liability. The resulting excess mass near the threshold can be compared with a counterfactual density to identify the behavioral response.

The modern bunching estimator originates with Saez (2010), who shows that self-employed taxpayers bunch at kinks because they can adjust reported income, while wage earners, whose income is third-party reported, do not. This establishes a broader principle: independent reporting by another party narrows the scope for misreporting. Kleven and Waseem (2013) extend the approach to notches using Pakistani data, showing that optimization frictions such as limited information and adjustment costs attenuate bunching relative to a frictionless benchmark. Kleven (2016) clarify that a kink shifts only the marginal rate, whereas a notch such as a VAT threshold shifts the average rate discontinuously, producing excess mass below the cutoff and missing mass above it.

Bunching estimates are sensitive to researcher choices such as polynomial degree and exclusion range (Kleven 2016), and to assumptions about unobserved heterogeneity (Blomquist et al. 2021; Bertanha, McCallum, and Seegert 2023; Anagol et al. 2024). I mitigate these concerns by embedding the estimates in a difference-in-differences framework that exploits sectoral variation over time, rather than treating any single bunching estimate as a structural parameter.

The threshold is also a policy choice, not merely an administrative parameter. Kanbur and Keen (2014) show that the avoidance opportunities a VAT notch creates can push the welfare-optimal threshold above the level implied by administrative costs alone. In low-capacity settings such as Mongolia, the threshold largely determines which firms can realistically be brought into the tax net, and so shapes the effective scope of enforcement.

Firms respond to the notch along several margins. Onji (2009) finds that large Japanese firms split into smaller units to stay below the cutoff, bunching by reorganization rather than misreporting or output restraint. Harju, Matikka, and Rauhanen (2019) find that Finnish firms react more strongly to compliance costs than to VAT rates, so administrative burden often dominates the tax incentive. Muthitacharoen, Wanichtha-

worn, and Burong (2021) document substantial bunching in Thailand and attribute much of it to income misreporting, and Liu et al. (2021) find comparable threshold bunching across firm types in the United Kingdom. The regularity is common across settings: firms bunch below the VAT threshold, and how hard they bunch turns as much on compliance costs and detection risk as on the rate itself.

VAT enforcement is weakest at the final sale, and the weakness is structural. B2B transactions leave a bilateral invoice trail that authorities can cross-check, whereas B2C transactions do not, since consumers neither file VAT returns nor claim input credits and so have little reason to demand receipts. Pomeranz (2015) provides experimental evidence that audit threats reduce underreporting in B2C transactions but have limited effect where B2B invoice trails already constrain evasion, and Almunia and Lopez-Rodriguez (2018) show that enforcement works better when transaction records are richer and easier to verify. This is why consumer exposure governs bunching intensity: a purely consumer-facing firm bears the full output VAT on registration, while a B2B firm’s customers reclaim it (Liu et al. 2021).

Governments have responded by extending third-party reporting to the consumer stage. Naritomi, Nyamdavaa, and Campbell (2025) survey such schemes across forty-four countries, and Naritomi (2019) shows that Brazil’s Nota Fiscal Paulista raised reported sales through consumer rebates and lotteries. Mongolia’s ebarimt combines a VAT rebate with a national lottery; Nyamdavaa (2023) documents significant increases in reported liabilities among B2C retailers after its launch, along with upstream spillovers to their wholesale suppliers. Its mechanics are detailed in Chapter 3.1.

The bunching and consumer-enforcement literatures have largely developed separately. The closest study is Nyamdavaa (2023), which estimates compliance responses to ebarimt with a difference-in-differences design but does not examine threshold bunching. This thesis connects the two by combining polynomial bunching estimates with a difference-in-differences framework that exploits cross-sector variation in B2C exposure.

Chapter 3

Methodology and Data

This chapter sets up the empirical analysis. Section 3.1 describes Mongolia’s VAT regime and the 2016 reforms. Section 3.2 develops the notch model that motivates the design and states the predictions it generates. Section 3.3 describes the binned tax data, and Section 3.4 sets out the bunching estimator and the difference-in-differences design.

3.1 Institutional Background

Mongolia introduced the Value-Added Tax (VAT) in July 1998 (Mongolian Tax Administration 2023). A flat rate of 10 percent applies to domestic sales of goods and services, while exports are zero-rated. Firms whose annual sales exceed the registration threshold must register, file monthly returns, and remit their net VAT liability. Firms below the threshold are exempt and cannot claim input tax credits.¹

The threshold was set at 10 million MNT in June 2001, about 9,100 USD at the time, and remained unchanged for nearly fifteen years. By 2015, inflation and currency depreciation had eroded its real value to roughly 5,000 USD, drawing a growing number of small firms into the VAT system. Over this period VAT accounted for between 17 and 33 percent of total tax revenue (National Statistics Office of Mongolia).

In January 2016, the revised VAT Law enacted two reforms at once. First, the registration threshold rose fivefold to 50 million MNT (about 23,000 USD), reducing compliance obligations for smaller firms and letting the authority concentrate enforcement on larger businesses. Second, the government launched ebarimt, an electronic receipt system aimed at strengthening compliance in retail transactions by drawing consumers into enforcement.

Ebarimt gives consumers a financial reason to request and register receipts. Those who upload receipts through the mobile application receive a rebate equal to 20 percent of the VAT paid and enter monthly lottery draws with prizes up to 20 million MNT

1. An unregistered firm therefore pays VAT on its own purchases without recovering it, the standard cost of remaining below the threshold.

(about 9,300 USD) (Mongolian Tax Administration 2023).² Consumers can also report firms that refuse to issue receipts directly through the application. On the firm side, businesses must install point-of-sale systems linked to the ebarimt network (Nyamdavaa 2023); once a receipt is issued, the transaction is transmitted to the tax authority regardless of whether the consumer later registers it. Reporting thus occurs at the point of sale rather than through subsequent consumer action, and non-compliant firms face administrative penalties.

The reform altered monitoring of B2C transactions specifically. Before ebarimt, consumers had little reason to request receipts, which limited the authority’s ability to verify retail sales; B2B transactions, by contrast, already generated verifiable records through VAT invoices and input-credit claims (Chapter 2). This asymmetry is the basis for using B2B sectors as the comparison group throughout the analysis, and Section 3.2 formalizes it.

3.2 Theoretical Framework

This section formalizes how a discrete tax notch induces revenue compression and how the asymmetric enforcement shock of Section 3.1 shifts that compression differentially across sectors.

Consider a continuum of firms indexed by productivity $n \in [n_L, n_H]$. Each firm chooses reported annual revenue z to maximize utility subject to the VAT threshold z_t^* , which equals $z_1^* = 10,000,000$ MNT (about 4,700 USD) before the reform (2014–2015) and $z_2^* = 50,000,000$ MNT (about 23,000 USD) after (2016–2023). In both regimes the schedule introduces a discrete notch:

$$T(z; z_t^*) = \begin{cases} 0 & \text{if } z < z_t^* \\ \tau z & \text{if } z \geq z_t^* \end{cases} \quad (3.1)$$

where $\tau = 0.10$ is the statutory rate. A firm crossing z_t^* faces a discrete liability jump of $\Delta T = \tau z_t^*$, creating an incentive to report just below the threshold (Kleven and Waseem 2013; Kleven 2016). Because ΔT scales with z_t^* , this mechanical incentive is five times larger after 2016 ($\Delta T_2 = 5,000,000$ MNT (about 2,300 USD)) than before ($\Delta T_1 = 1,000,000$ MNT (about 470 USD)).

Modeling the levy as τz on gross revenue is exact for B2C sales: with no downstream credit, the VAT is a turnover tax, so the gain from staying unregistered is largest for consumer-facing firms (Liu et al. 2021).³ For B2B sectors the incentive is weaker, since

2. If no top-prize winner is identified, the prize rolls over, up to a maximum jackpot of 500 million MNT (about 232,000 USD).

3. Naritomi (2019) models tax as proportional to gross reported revenue and notes the logic extends to VAT when reported expenses do not fully offset reported revenue; Harju, Matikka, and Rauhanen

registered buyers reclaim the seller's output VAT. This is the identification lever: B2B sectors supply the counterfactual trend rather than the object of inference.

A proportional notch generates a dominated region just above the threshold (Kleven and Waseem 2013; Kleven 2016), though heterogeneous input shares blur its upper edge (Harju, Matikka, and Rauhanen 2019; Liu et al. 2021). Recovering the structural elasticity ϵ from the marginal-buncher condition requires parametric assumptions that cannot be credibly met with binned, sector-level data (Blomquist et al. 2021). I therefore do not recover ϵ and treat the reduced-form bunching ratio as the primary estimand:

$$\hat{b}_{st} = \frac{\hat{B}_{st}}{\hat{N}_{R_t}} \quad (3.2)$$

where $\hat{B}_{st}$ is the excess firm count in the bunching window and $\hat{N}_{R_t}$ the average counterfactual count over it; normalizing by the local counterfactual height removes the mechanical thinning of the firm distribution at higher revenue.

Where transactions are hard to verify and cash is common, firms reach the threshold by under-reporting rather than by cutting real output. Following Allingham and Sandmo (1972), let y be true revenue and $p(s, t) \in (0, 1)$ the sector-time audit probability. A firm with $y > z_t^*$ under-reports to $z \approx z_t^*$ and bunches when the expected profit from doing so exceeds honest compliance:

$$\underbrace{y}_{\text{Untaxed at } z_t^*} - p(s, t)[(\tau + \theta)(y - z_t^*)] - c(y - z_t^*) > \underbrace{y(1 - \tau)}_{\text{Honest compliance}} \quad (3.3)$$

where $\theta > 0$ is a proportional penalty on concealed revenue and $c(\cdot)$ a strictly convex friction cost with $c(0) = 0$.⁴ The condition defines a productivity threshold below which firms manipulate, and the excess mass widens as the evasion zone widens. Differentiating the boundary condition gives the two comparative statics that drive the empirical work:

$$\frac{\partial \hat{b}_{st}}{\partial z_t^*} > 0 \quad \text{and} \quad \frac{\partial \hat{b}_{st}}{\partial p(s, t)} < 0. \quad (3.4)$$

A higher threshold raises bunching by enlarging the untaxed allowance and the liability jump; a higher detection probability lowers it by raising the expected cost of concealment. Nothing in the empirical design depends on the level of ϵ ; the model is used only for the signs in equation (3.4) and the three predictions that follow.

The 2016 reform moves both arguments at once. The threshold rises from 10M to 50M for all sectors, pushing $\hat{b}$ up everywhere. Ebarimt adds an enforcement shock $\Delta p > 0$ that

(2019) model the notch as a levy on value added, with the incentive scaling in the value-added share.

4. If detected with probability $p(s, t)$, the authority recovers true revenue y and assesses the omitted VAT $\tau(y - z_t^*)$ plus a fine $\theta(y - z_t^*)$, following Allingham–Sandmo (1972) and Naritomi (2019).

reaches only consumer-facing sectors through POS-linked third-party reporting, so that $p(\text{B2C}, t \geq 2016) = p(\text{B2C}, t < 2016) + \Delta p$ while $p(\text{B2B}, t \geq 2016) \approx p(\text{B2B}, t < 2016)$.⁵ Combined with the pre-reform ordering $p(\text{B2C}, t < 2016) \ll p(\text{B2B}, t)$, equation (3.4) yields three predictions:

1. *Pre-reform gap.* Weaker baseline B2C enforcement implies higher baseline bunching in B2C than B2B sectors at the 10M threshold.
2. *No secular B2B trend.* Facing no consumer-side shock, B2B excess mass should show no systematic post-reform decline, tracking the macro trajectory and the larger threshold and so providing the counterfactual for the pure threshold shift.⁶
3. *Negative DiD estimand.* Because Δp offsets the threshold-induced rise more in B2C than B2B sectors,

$$\beta = (\hat{b}_{\text{B2C, Post}} - \hat{b}_{\text{B2C, Pre}}) - (\hat{b}_{\text{B2B, Post}} - \hat{b}_{\text{B2B, Pre}}) < 0. \quad (3.5)$$

3.3 Data

I use administrative Corporate Income Tax (CIT) data from the Mongolian Tax Authority. Although the notches govern VAT registration, the CIT registry has an advantage for bunching estimation: because all incorporated entities must file income tax returns regardless of VAT status, the revenue distribution spans both sides of the threshold and avoids the survival selection inherent in VAT-only registries.

To protect confidentiality while preserving behavioral variation near the notches, the microdata are aggregated into a balanced panel of binned revenue frequencies. Gross annual revenues are sorted into uniform 1,000,000 MNT (about 470 USD) bins from 0 to 400,000,000 MNT (about 186,000 USD), giving 400 bins per cross-section. The panel covers four sectors over 2014–2023, for 40 sector-year cells.

This format rules out firm-level covariates and individual robustness checks common in microdata studies (Harju, Matikka, and Rauhanen 2019; Liu et al. 2021; Muthitacharoen, Wanichthaworn, and Burong 2021), but it suits the estimator, which operates directly on bin frequency counts (Kleven 2016), and the 400 bins per cell leave ample room to fit flexible polynomial counterfactuals.

The sample splits into two blocks by consumer exposure. The B2C treatment group comprises retail trade and food services, which face final consumers and lacked third-party sales trails before 2016. The B2B control group comprises wholesale trade and

5. E-receipts transmit automatically once the POS terminal is online, with at most three working days' delay (Nyamdavaa 2023). Rollout was gradual, about 51 percent of retailers enrolled by 2016 and 56 percent by 2018, so Δp is an average shock across the treated sector, not a uniform jump.

6. This is partially violated if ebarimt raised B2B compliance through invoice-matching pressure (Nyamdavaa 2023); Section 4.3 addresses this by excluding Wholesale Trade from the control group.

manufacturing, which operate on inter-firm lines where invoice matching and deduction incentives already suppressed evasion (Pomeranz 2015). Because the lottery and rebate work through the consumer-sales margin, they leave B2B registration incentives largely intact. Any supply-chain spillover would compress control-group bunching too and attenuate $\hat{\beta}$ toward zero, a concern tested in Section 4.3.

Table 3.1: **Aggregate Firm Distribution by Sector Group and Policy Regime**

Sector	<i>Pre-Reform (2014–2015)</i>		<i>Post-Reform (2016–2023)</i>	
	Mean Firms ^a	Annual Total ^b	Mean Firms ^a	Annual Total ^b
<i>B2C Treatment Group</i>				
Food service	5.3	1,178	4.3	1,231
Retail	20.1	7,228	20.0	7,890
<i>B2B Control Group</i>				
Manufacturing	8.9	3,015	9.3	3,393
Wholesale	19.2	7,519	23.9	9,564

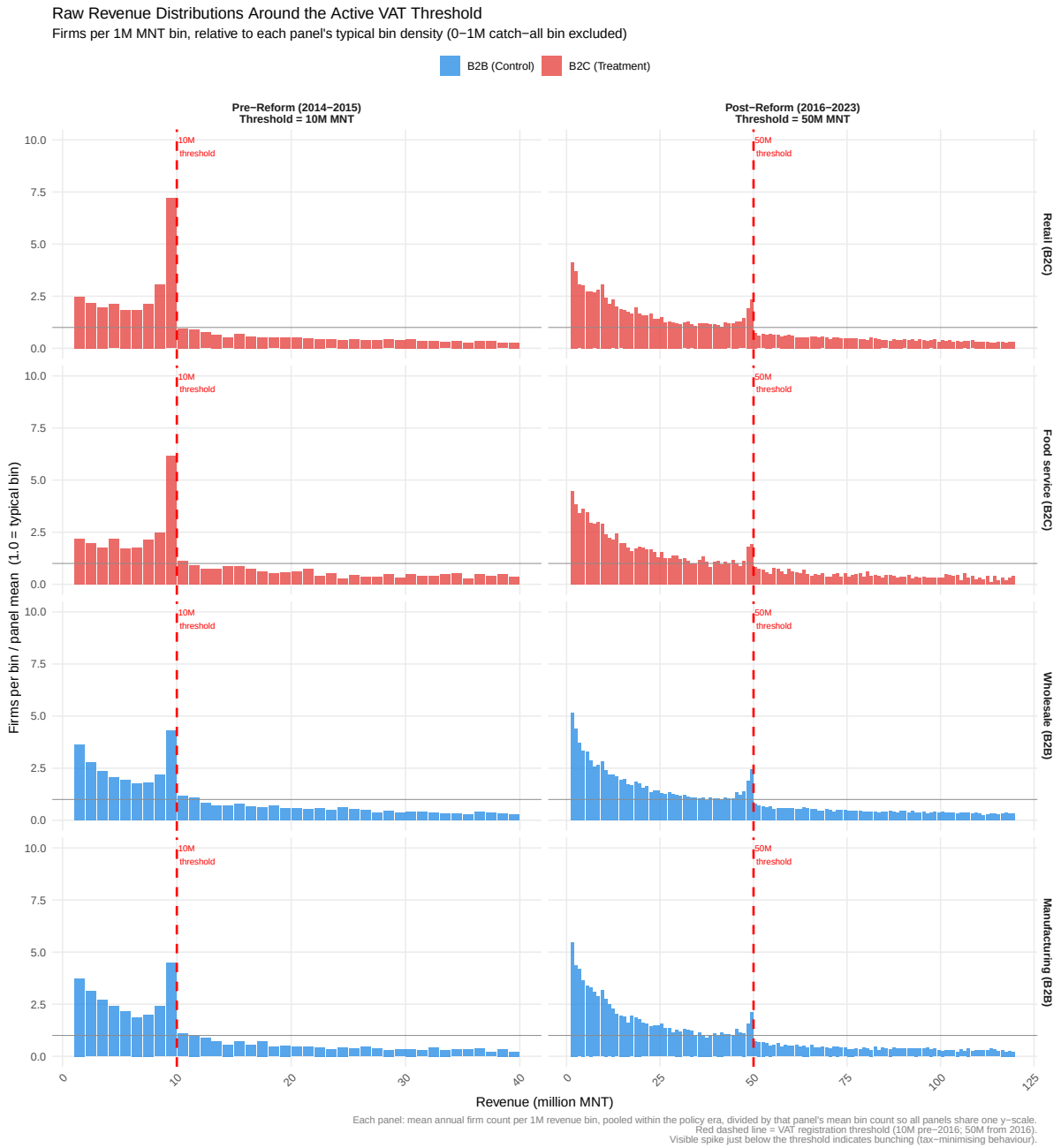
Notes: ^a Average annual firms per active revenue bin across 400 bins (0–400M MNT).

^b Mean annual firm count excluding X-filers. Source: MTA CIT registry.

Table 3.1 reports firm distributions by sector and period. Before the reform, retail and wholesale were comparable in size (7,228 and 7,519 firms per year), which supports wholesale as a comparison group. The largest post-reform change is in wholesale, up 24.5 percent (19.2 to 23.9 firms per bin), consistent with the upstream spillover of Nyamdavaa (2023): as retail sales became more visible, suppliers may have faced more pressure to report accurately. The implications for $\hat{\beta}$ are taken up in Sections 4.2 and 4.3. Food services declines modestly (5.3 to 4.3), most likely reflecting redistribution of firms after the fivefold threshold increase rather than a real contraction.

The density discontinuity at the threshold is large (Appendix Table A.1). In 2014 the bin just below $z_1^* = 10,000,000$ MNT (about 470 USD) holds 1,105 retail firms against 98 just above, an 11.3:1 ratio. The B2B ratios are far lower (4.7:1 wholesale; 5.6:1 manufacturing), direct descriptive evidence of the B2C–B2B contrast and a first indication consistent with Prediction 1. By 2017 the B2C ratios compress (retail 3.3:1; food services 3.2:1) while the B2B ratios fall more modestly (wholesale 2.8:1; manufacturing 1.9:1). Figure 3.1 shows the pattern.

Figure 3.1: Raw Revenue Distributions by Sector and Policy Era



Notes: Mean annual firms per 1 million MNT revenue bin by policy period. The [0,1M) MNT bin is excluded. Red dashed line marks the VAT threshold. *Source:* MTA CIT registry.

To protect parallel trends, the estimation samples vary in their treatment of the pandemic years, when containment policy hit the B2C and B2B groups asymmetrically. I define three nested samples:

1. **Main (preferred):** 36 cells, 2014–2019 and 2021–2023, omitting 2020.
2. **Robustness Sample A:** 32 cells, 2014–2019 and 2022–2023, omitting 2020 and 2021.

3. **Full Sample:** all 40 cells, 2014–2023, retained as a conservative benchmark.

The 2020 exclusion reflects Mongolia’s zero-COVID measures, which closed in-person retail and food service (B2C) while freight exemptions kept wholesale and manufacturing (B2B) operating; the further 2021 exclusion in Sample A reflects extended Ulaanbaatar lockdowns. Reporting all three samples shows how far the estimate moves with the pandemic years, which Section 4.3 quantifies.

3.4 Estimation Method

Estimation proceeds in two steps. First, for each sector-year I recover a normalised bunching ratio $\hat{b}_{st}$ using the polynomial counterfactual method of Chetty et al. (2011), adapted to the notch setting following Kleven and Waseem (2013), Harju, Matikka, and Rauhanen (2019), and Liu et al. (2021). Second, I use the recovered $\hat{b}_{st}$ as the outcome in a difference-in-differences design comparing B2C and B2B sectors around 2016.

The bunching estimate. Let N_{jst} be the firm count in bin j , sector s , year t , with midpoint z_j , and let

$$z_t^* = \begin{cases} 10,000,000 \text{ MNT} & t \in \{2014, 2015\} \\ 50,000,000 \text{ MNT} & t \in \{2016, \dots, 2023\}. \end{cases} \quad (3.6)$$

The smooth counterfactual $\hat{N}_{jst}$ is recovered in three steps.

1. *Step 1: counterfactual fit.* A polynomial centred at the threshold is fitted to the observed distribution, with dummies absorbing the bins inside an exclusion window W_t ,

$$N_{jst} = \sum_{k=0}^p \gamma_k (z_j - z_t^*)^k + \sum_{l \in W_t} \nu_l \mathbb{I}(j = l) + u_{jst}, \quad (3.7)$$

with $p = 5$ pre-reform and $p = 7$ post-reform. The counterfactual is the polynomial prediction with the window dummies omitted, $\hat{N}_{jst} = \sum_{k=0}^p \hat{\gamma}_k (z_j - z_t^*)^k$, floored at zero. Since the dominated region scales with the threshold at a fixed rate (Kleven and Waseem 2013), a window fixed in absolute MNT would be too narrow post-reform, so the windows scale with z_t^* :

$$W_t = \begin{cases} [z_1^* - 2\text{M}, z_1^* + 10\text{M}) = [8\text{M}, 20\text{M}) & t \in \{2014, 2015\} \\ [z_2^* - 2\text{M}, z_2^* + 30\text{M}) = [48\text{M}, 80\text{M}) & t \in \{2016, \dots, 2023\}. \end{cases} \quad (3.8)$$

The lower edge sits 2M below the threshold in both regimes, capturing the spike just below z_t^* without reaching the rising shoulder of the small-firm density (Figure 3.1);

the upper edge is the maximum extent of the Step 2 search. Sensitivity to the window appears in Appendix Table B.3.

2. *Step 2: upper bound.* The upper bound z_t^U cannot be read directly from the data, since missing mass above a notch is diffuse. Following Kleven and Waseem (2013), it is pinned down by setting missing mass equal to bunching mass,

$$\hat{M}_{st} = \hat{B}_{st}, \quad \hat{M}_{st} = \sum_{j \in [z_t^*, z_t^U]} (\hat{N}_{jst} - N_{jst}), \quad (3.9)$$

raising z_t^U from z_t^* in 1M increments, re-estimating the counterfactual at each step until equation (3.9) first holds, capped at the upper edge of W_t . Where the condition is met within the cap, it holds up to the 1M increment; in cells with no clean upper hole it has no interior solution, and z_t^U is set to the cap and the cell flagged.

3. *Step 3: bunching ratio.* The bunching window is $R_t = [z_t^* - 2,000,000, z_t^*)$, narrower than W_t by design: W_t must reach far enough from the threshold to fit the counterfactual on unaffected bins, whereas bunching is a near-point mass in the bin just below z_t^* . Bins farther down carry near-zero excess mass, so widening R_t would dilute $\hat{b}_{st}$; stability to alternative widths is confirmed in Appendix Table B.3. The excess frequency is $\hat{B}_{st} = \sum_{j \in R_t} (N_{jst} - \hat{N}_{jst})$, and

$$\hat{b}_{st} = \frac{\hat{B}_{st}}{\hat{N}_{R_t}}, \quad \hat{N}_{R_t} = \frac{1}{|R_t|} \sum_{j \in R_t} \hat{N}_{jst}. \quad (3.10)$$

Standard errors for $\hat{b}_{st}$ come from a residual bootstrap with 500 replications within each cell (Chetty et al. 2011): I resample the residuals of equation (3.7), regenerate bin counts, and re-estimate $\hat{b}_{st}$, holding z_t^U fixed across replications. Because $\hat{b}_{st}$ is a reduced-form outcome rather than an input to a structural elasticity, I follow Harju, Matikka, and Rauhanen (2019) and Liu et al. (2021) in not converting it to an elasticity; for a VAT notch, heterogeneous input costs prevent a cleanly defined dominated region above the threshold.

The difference-in-differences estimate. The recovered ratios enter a two-step difference-in-differences design,

$$\hat{b}_{st} = \alpha + \beta (\text{Post}_t \times \text{Treated}_s) + \gamma_s + \delta_t + \eta_{st}, \quad (3.11)$$

where $\text{Post}_t = \mathbb{I}(t \geq 2016)$ and $\text{Treated}_s = 1$ for the B2C sectors (retail and food service), 0 for the B2B sectors (wholesale and manufacturing). Sector effects γ_s absorb time-invariant industry differences; year effects δ_t absorb common shocks and the economy-

wide level shift in $\hat{b}$ from the threshold increase. Because that increase applies to all sectors by statute, the differential B2C–B2B change in $\hat{b}_{st}$ is attributable to ebarimt to the extent that the threshold increase shifts both groups equally; under that common-threshold assumption, β identifies the effect of consumer monitoring.⁷

Two features of the design warrant comment. First, $\hat{b}_{st}$ enters the second stage as an estimated rather than observed outcome, a generated-dependent-variable structure. Second, the two-step approach is deliberate: Arkhangelsky, Yanagimoto, and Zohar (2024) show that one-step alternatives, which pool the microdata and interact the policy directly, can be inconsistent because the treatment shifts the implicit estimation weights, whereas separating parameter estimation from policy evaluation avoids this. The within-cell residual bootstrap quantifies the first-stage error (Appendix Table A.3); the second stage treats $\hat{b}_{st}$ as given and does not propagate that error jointly. The reported second-stage standard errors are therefore lower bounds, so the true p -values lie somewhat above those in Table 4.2, a simplification revisited in Chapter 5.

With only $G = 4$ clusters, asymptotic cluster-robust variance estimation over-rejects severely (Cameron, Gelbach, and Miller 2008; MacKinnon, Nielsen, and Webb 2023). I therefore base inference on the restricted wild cluster bootstrap, imposing the null when generating bootstrap residuals (MacKinnon, Nielsen, and Webb 2023), with Webb six-point weights: at $G = 4$, Rademacher weights admit only $2^4 = 16$ sign vectors and a minimum two-sided p -value of 0.125, while Webb weights give $6^4 = 1,296$ distinct samples and finer resolution. I draw $B = 9,999$ replications. Inference with four clusters remains limited, so I read the bootstrap p -value as suggestive and corroborate it with the stability of $\hat{\beta}$ across samples and with the event-study evidence below.

To assess parallel trends and trace timing, I estimate

$$\hat{b}_{st} = \alpha + \sum_{t \neq 2015} \beta_t (\mathbb{I}(\text{year} = t) \times \text{Treated}_s) + \gamma_s + \delta_t + \eta_{st}. \quad (3.12)$$

The baseline year is $t_0 = 2015$, the last pre-reform year, which predates any anticipatory response to the threshold change.⁸ Pre-2016 estimates test parallel trends; post-reform estimates trace whether the response is immediate or accumulates with ebarimt coverage.

I report the three samples of Section 3.3, together with Robustness Sample B, which drops wholesale trade from the control group to address possible ebarimt spillovers through supply linkages. Section 4.3 examines how far the estimate moves with the pandemic years and with the spillover concern.

7. β is not collinear with δ_t : the year effects absorb the common annual level of $\hat{b}$ across both groups, while β identifies only the treated-minus-control deviation.

8. Results are robust to omitting $t_0 = 2014$ instead; see Section 4.3.

Chapter 4

Estimation Results and Empirical Analysis

The results proceed in three steps: the bunching estimates that motivate the causal analysis (Section 4.1), the difference-in-differences estimate and its inference (Section 4.2), and the dynamics and credibility checks (Section 4.3).

4.1 Non-Parametric Bunching Estimates and Baseline Density Analysis

The polynomial counterfactual of Section 3.4 is applied to each of the 36 sector-year cells in the main sample, yielding the normalised ratio $\hat{b}_{st}$ and its bootstrap standard error. Figure A.1 shows the full 2014 and 2017 distributions for all four sectors; the 2014 retail density drop across z_1^* documented in Section 3.3 is the most pronounced raw case. Table 4.1 reports the group means, and sector-level estimates for all 36 cells appear in Appendix Table A.3.

Table 4.1: **Normalised Bunching Estimates: Group Means and Enforcement Gap (Main Sample, 2020 Excluded)**

Structural Metric	2014	2015	2016	2017	2018	2019	2021	2022	2023
B2C mean	4.156	3.230	2.185	1.667	1.837	2.392	1.026	3.009	1.644
B2B mean	2.660	1.844	1.471	2.340	2.841	2.000	1.330	2.056	2.570
Gap (B2C – B2B)	+1.497	+1.386	+0.715	–0.673	–1.004	+0.393	–0.304	+0.953	–0.926

Notes: Group means of $\hat{b}_{st}$ estimated via a 7-degree polynomial counterfactual (excl. window $[z_t^* - 2M, z_t^* + 30M)$ post-reform). B2C = retail trade + food services; B2B = wholesale + manufacturing. 2020 excluded. *Source:* Author’s calculations based on MTA CIT registry data.

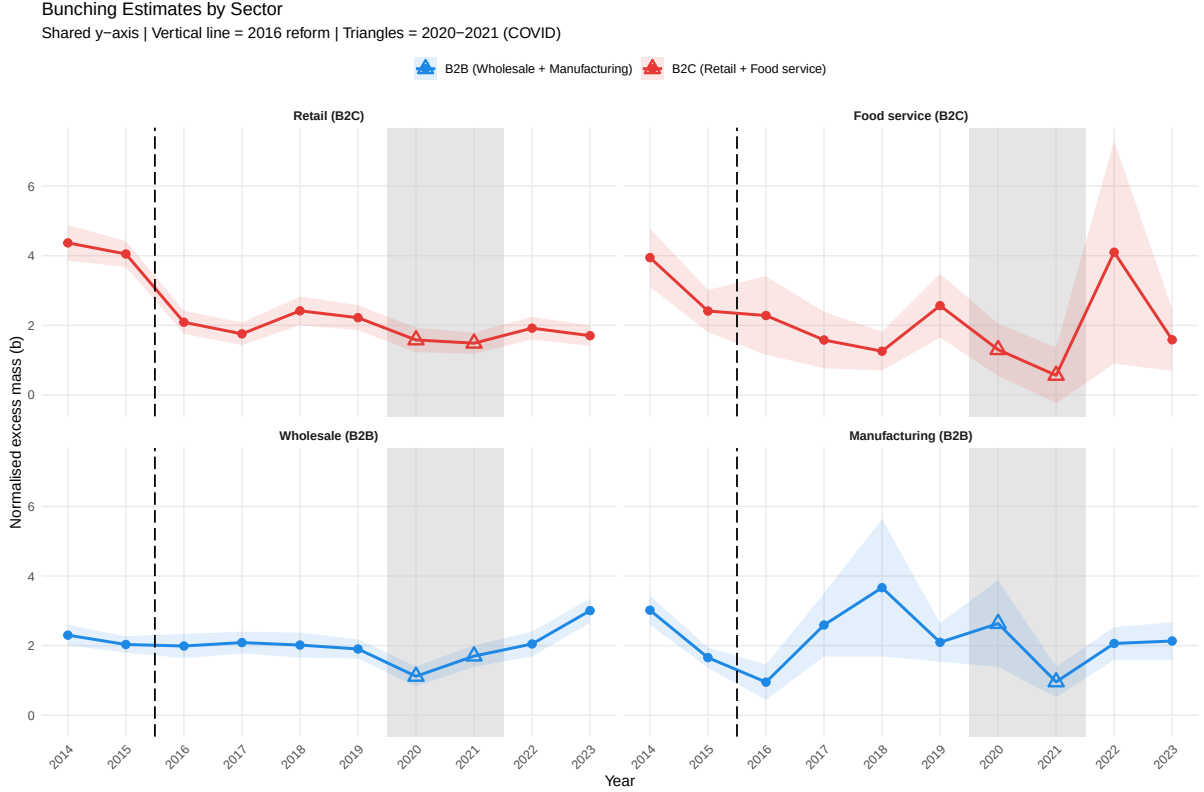
Before the reform, B2C sectors bunch far harder than B2B: a mean $\hat{b}$ of 3.693 against

2.252, a baseline gap of 1.441 standardised units. This matches Prediction 1 and the breakdown of VAT self-enforcement at the final consumer stage (Pomeranz 2015): without a downstream buyer demanding receipts, firms selling to households face lower detection costs and sustain higher bunching. The gap is widest in 2014, when retail records $\hat{b} = 4.368$ (SE 0.263) and food service $\hat{b} = 3.945$ (SE 0.430; Table A.3). Most pre-reform cells are flagged in Table A.3: the diffuse upper hole characteristic of a VAT notch, where heterogeneous input shares blur the missing-mass region (Harju, Matikka, and Rauhanen 2019; Liu et al. 2021), leaves estimated missing mass short of bunching mass, so z_t^U is capped at the window edge. This affects the upper-bound search rather than the sub-threshold excess that defines $\hat{b}_{st}$, and the raw density discontinuities in Table A.1 (11.3:1 in retail against 4.7:1 in wholesale, 2014) show the same B2C–B2B gap with no counterfactual at all.

After the reform the gap narrows and reverses. It turns negative by 2017 (-0.673), as B2C $\hat{b}$ falls below the B2B control, and reaches -0.926 by 2023 (B2C 1.644 against B2B 2.570). The year-to-year path is not monotone, with a temporary 2022 B2C spike driven by an anomalous food service estimate, but the post-reform gap sits well below its pre-reform level. Relative to B2B controls, B2C threshold distortions fell far enough to eliminate the pre-existing gap and reverse its sign.

The B2B control drifts upward over the post-reform period, most visibly in wholesale ($\hat{b} = 3.006$ in 2023, up from 2.033 in 2015). Section 4.2 returns to this in the contamination caveat. One plausible contributor is mechanical: the fivefold increase in z^* may have moved more B2B firms into the neighbourhood of the active notch, raising the scope for clustering without any change in enforcement.

Figure 4.1: Normalised Bunching Ratio $\hat{b}_{st}$ by Sector, 2014–2023



Notes: Sector-specific $\hat{b}_{st}$ estimates with 95% bootstrap confidence intervals (shaded). The dashed line marks the January 2016 VAT reform. Triangles indicate 2020 and 2021, the years affected by the COVID-19 pandemic. *Source:* Author's calculations based on MTA CIT registry data.

Figure 4.1 plots the four sectors separately, and the clearest contrast is between food service and manufacturing (sector-year values in Table A.3). Food service shows the sharpest pre-reform decline of any sector, from 3.945 in 2014 to 2.411 in 2015, and continues down through the post-reform period to 0.564 in 2021, implying little excess mass below the new 50M MNT threshold. This pattern is what the mechanism of Section 3.2 would produce in the sector most exposed to foot traffic and receipt scanning: a rise in $p(\text{B2C}, t \geq 2016)$ removes the expected gain from evasion for all but the most productive firms, and the bunching mass collapses with it. The pattern echoes Naritomi (2019), who finds the largest consumer-receipt compliance gains among smaller B2C firms, where the fixed cost of suppressing receipts is highest relative to the tax at stake. The 2022 food service reading ($\hat{b} = 4.099$) is an outlier from sparse near-threshold bins and an ill-conditioned counterfactual; it does not survive the polynomial-degree checks in Appendix Table B.2.

Manufacturing, by contrast, holds broadly stable through 2019, dips in 2021 ($\hat{b} = 0.959$), and recovers to 2.063 (2022) and 2.133 (2023). It shares only the 2021 trough with food service; otherwise the paths diverge in what matters. Food service declines

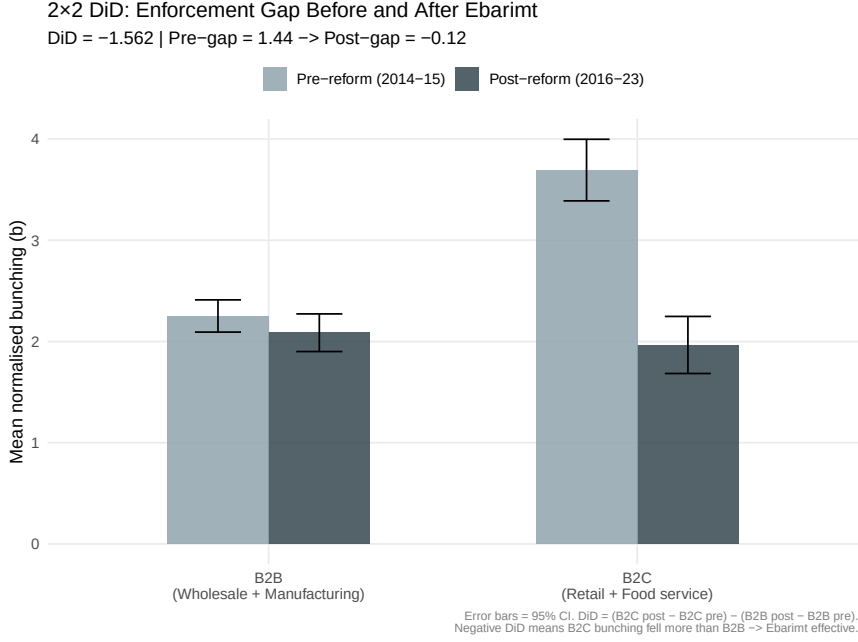
before the reform and settles well below its pre-reform level, whereas manufacturing has no secular decline and returns to its pre-reform level by 2023. The 2021 dip is consistent with COVID-era supply disruption rather than a compliance shift, as Section 4.3 examines directly. With no exposure to consumer monitoring, manufacturing bunching tracks the macroeconomy and the structure of the new notch, which is the comparison the design requires.

Figures A.6 and A.7 present the counterfactual overlays for retail and food service in 2015 and 2022, bracketing the reform. The fits track the observed density smoothly away from the exclusion window, and the predicted holes above the threshold match the missing mass implied by the notch, so the estimated excess mass is not an artefact of window selection. The overlays also fix the magnitude: in 2015 the retail bunching window holds an excess of roughly 768 firms ($\hat{b} = 4.05$) above the counterfactual; by 2022 that excess falls to about 100 firms ($\hat{b} = 1.92$), an 87% reduction in raw terms. Food service moves the same way over most of the window, with $\hat{b}$ falling from 2.41 in 2015 toward its floor by 2021 (the elevated 2022 reading is the artefact noted above and is set aside).

4.2 Primary Difference-in-Differences Estimation Results

The sector-year ratios from Table 4.1 enter the two-step DiD framework of Equation (3.11). Figure 4.2 shows the 2×2 decomposition: mean $\hat{b}$ for the B2C and B2B groups across the pre- and post-reform periods.

Figure 4.2: **2×2 DiD: Mean Normalised Bunching Before and After the 2016 Reform**



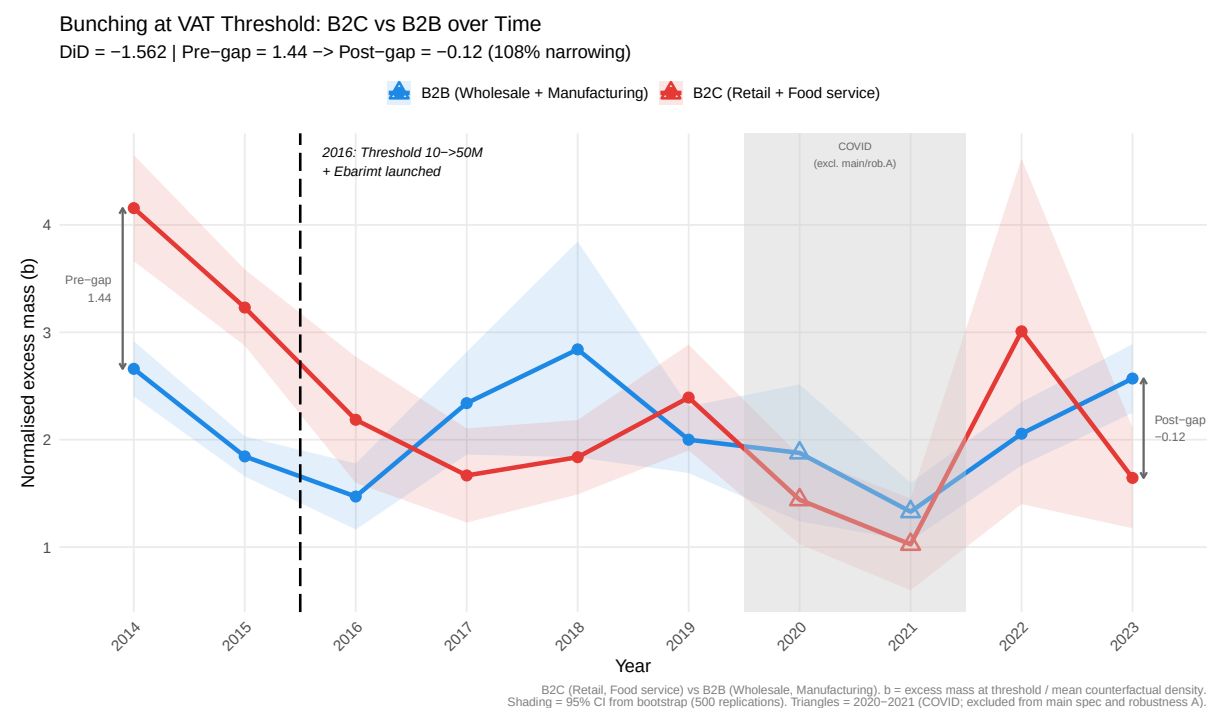
Notes: Group means of $\hat{b}_{st}$ by treatment group and policy regime. Error bars denote 95% confidence intervals. Pre-reform period: 2014–2015; post-reform period: 2016–2023, excluding 2020. The baseline DiD estimate ($\hat{\beta} = -1.562$) is from the Main specification (Table 4.2, column 1). *Source:* Author’s calculations based on MTA CIT registry data.

The decomposition is direct (Table 4.2, column 1). B2C mean $\hat{b}$ falls from 3.693 pre-reform to 1.966 post-reform, a within-group decline of 1.727 units, while the B2B control falls only from 2.252 to 2.087, a change of -0.165 . Under parallel trends, the B2B trajectory supplies the counterfactual: the threshold shift alone would have reduced B2C bunching by about 0.165 units, so the residual reduction attributable to the differential enforcement shock is $1.727 - 0.165 = 1.562 = -\hat{\beta}$. The regression confirms this:

$$\hat{\beta} = -1.562. \quad (4.1)$$

This compresses the pre-reform enforcement gap from +1.441 to -0.121 standardised units, eliminating the original differential and flipping its sign: after the reform B2C sectors bunch less than B2B sectors, a reversal hard to explain without a real change in how consumer-stage sales are verified. Figure 4.3 traces this convergence over the full sample period.

Figure 4.3: **B2C vs. B2B Normalised Bunching Over Time and the Convergence of the Enforcement Gap**



Notes: Pooled B2C and B2B mean $\hat{b}_{st}$ series with 95% bootstrap confidence intervals (shaded). The dashed line marks the January 2016 reform. The benchmark DiD estimate ($\hat{\beta} = -1.562$) implies a 108% reduction in the initial B2C–B2B gap. Triangles indicate the COVID-19 years. *Source:* Author's calculations based on MTA CIT registry data.

Table 4.2: **Difference-in-Differences Estimation Results: Three Samples**

	(1) Main (2020 excl.)	(2) Rob. A (2020+21 excl.)	(3) Full Sample (all years)
$\hat{\beta}$: Post $\times$ Treated	-1.562* (0.554)	-1.532* (0.633)	-1.602* (0.519)
Webb p -value	0.066	0.079	0.060
Pre-reform B2C mean ($\bar{\hat{b}}$)	3.693	3.693	3.693
Post-reform B2C mean ($\bar{\hat{b}}$)	1.966	2.122	1.900
Pre-reform B2B mean ($\bar{\hat{b}}$)	2.252	2.252	2.252
Post-reform B2B mean ($\bar{\hat{b}}$)	2.087	2.213	2.060
Pre-reform enforcement gap	+1.441	+1.441	+1.441
Post-reform enforcement gap	-0.121	-0.090	-0.160
Gap compression	108%	106%	111%
Sector FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
2020 excluded	Yes	Yes	No
2021 excluded	No	Yes	No
N (cells)	36	32	40

Notes: Dependent variable: normalised bunching ratio $\hat{b}_{st}$. CR1 cluster-robust standard errors (clustered by sector, $G = 4$) in parentheses. Significance stars are based on Webb wild cluster bootstrap inference (Webb 2023) with 6-point weights ($B = 9,999$ replications): * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Exclusion window: $[z_t^* - 2M, z_t^* + 30M)$ post-reform. *Source:* Author's calculations based on MTA CIT registry data.

The sign and magnitude of $\hat{\beta}$ match Prediction 3 (Section 3.2). In the model, ebarimt raises $p(\text{B2C}, t \geq 2016)$ by $\Delta p > 0$, tightening the evasion margin through $\partial \hat{b}_{st} / \partial p(s, t) < 0$ (Equation (3.4)). On both counts the decline is more consistent with enforcement than with mechanical displacement, and any control-group contamination would bias $\hat{\beta}$ toward zero.¹

The threshold-expansion confound. The 2016 reform bundled the fivefold threshold expansion with the ebarimt launch. Year fixed effects absorb the common response to the expansion, but if B2C and B2B sectors differed in firm-size distribution, the expansion alone could in principle produce a negative DiD coefficient. Two features of the data

1. The nominally fixed post-reform threshold $z_2^* = 50,000,000$ MNT (about 23,000 USD) loses real value as inflation erodes it over 2016–2023. This does not confound $\hat{\beta}$: symmetric inflation affects all sectors and is absorbed by the year fixed effects δ_t , while $\hat{\beta}$ is identified solely from differential B2C–B2B movement within each calendar year.

weigh against this. The timing does not fit a mechanical-displacement story: that would drop B2C $\hat{b}$ at once in 2016, whereas the 2016 event-study coefficient is indistinguishable from zero and the differential widens only later (Section 4.3). And the expansion should push $\hat{b}$ up for both groups, which is what the B2B control does (it rises, not falls); for the same expansion to also pull B2C down it would have to act in opposite directions on two groups facing the same notch. Section 4.3 develops the exogeneity of the reform more fully, including the compositional evidence from the threshold-zone data (Appendix Table A.2); on both counts here the decline is more consistent with enforcement than with mechanical displacement, and any control-group contamination would bias $\hat{\beta}$ toward zero.

Inference uses the restricted Webb wild cluster bootstrap of Section 3.4, appropriate at $G = 4$ clusters. The Main coefficient $\hat{\beta} = -1.562$ has Webb $p = 0.066$; the Full sample gives -1.602 ($p = 0.060$) and Robustness A -1.532 ($p = 0.079$). With four clusters the bootstrap cannot resolve finer significance, so the case rests on the stability of the point estimate, which moves only 0.07 units across specifications, and on the event-study pattern below rather than on significance stars.

4.3 Identification, Dynamics, and Robustness

A credible reading of $\hat{\beta}$ requires that it reflect enforcement rather than the exogeneity of the reform, divergent pre-trends, the treatment of pandemic years, or contamination of the control group. This section takes these in turn: it first sets out why the 2016 reform can be treated as exogenous to the differential margin the design exploits, then uses the event study to test parallel trends and trace the post-reform path, varies the sample to gauge sensitivity to the COVID years, and narrows the control group to check for spillover.

Exogeneity of the reform. The design treats the January 2016 reform as exogenous to the differential B2C–B2B margin it exploits. Two features of the setting support that reading, and a third, compositional, threat is testable and tested here.

The reform was not a response to sector-specific bunching. The threshold increase and the ebarimt launch were enacted together in the revised VAT Law, and I do not claim they are legislatively independent; both share a single origin. What identification requires is weaker. It is that the reform did not target the differential margin the design uses. The threshold applies to every sector by the same statutory rule, and nothing in the law conditions the cutoff or the receipt system on a sector’s prior evasion. The enforcement shock reaches consumer-facing firms not by legislative design but through the technology of the receipt: ebarimt binds where sales are made to final consumers who can be induced to demand receipts, and is close to inert where transactions are already invoiced between

firms (Section 3.2).

The threshold increase itself is best read as a broad administrative adjustment rather than a consumer-enforcement measure, which is what allows the year fixed effects to absorb it. The 10 million MNT cutoff had been fixed since 2001, and indexing it to consumer prices over the intervening fifteen years would imply a 2016 figure of roughly 37 million MNT.² The actual cutoff of 50 million MNT is of the same order, so the increase largely restored the threshold’s real value and eased filing for small firms, a motive common to B2C and B2B alike. Because that motive is sector-neutral, the year effects absorb it, and only the differential response is left for ebarimt to explain.

The residual threat is compositional rather than legislative, and the data speak to it directly. If B2C and B2B sectors differed systematically in firm-size distribution near the notch, the common threshold shift could move the two groups by different amounts even absent any enforcement change. The threshold-zone composition weighs against this: B2C and B2B firm counts sit near parity in fixed windows around both the old and the new notch before the reform (ratios of 1.13 and 0.88 in 2014), and both groups expand into the new threshold zone at comparable rates afterwards (Appendix Table A.2). The one systematic post-reform shift runs with the mechanism rather than against it: the B2C/B2B ratio at the old 10M notch falls from 1.09 in 2015 to 0.61 by 2023 as consumer-facing firms vacate that neighbourhood. The binned data cannot compare the full revenue distributions of threshold-adjacent firms across sectors, so compositional divergence cannot be excluded outright, a limit taken up in Chapter 5; the available evidence points to size-exposure symmetry. The event study adds a temporal check developed next: a composition-driven artefact of the threshold move would appear at once in 2016, whereas the differential emerges only as ebarimt coverage spreads.

With 2015 as the omitted base, the single pre-reform coefficient $\hat{\beta}_{2014} = +0.110$ ($p = 0.913$) is the parallel-trends test, and it is indistinguishable from zero, so the data give no evidence against parallel trends. Table 4.3 and Figure 4.4 report the full sequence; Robustness A yields identical coefficients for all non-pandemic years.

2. Author’s calculation cumulating annual consumer-price inflation from 2001 to 2016 (National Statistics Office). The point is only that the increase is of the same order as inflation catch-up, not a precise indexation.

Table 4.3: **Event Study Coefficients: B2C Differential Bunching Relative to the 2015 Reference Baseline**

Year	Main Specification (2020 Omitted)		Robustness A (2020 & 2021 Omitted)		Full Sample (All Years)	
	$\hat{\beta}_t$	p -value	$\hat{\beta}_t$	p -value	$\hat{\beta}_t$	p -value
<i>Pre-Reform Horizon</i>						
2014	+0.110	0.913	+0.110	0.915	+0.110	0.913
2015	0.000 (Baseline)		0.000 (Baseline)		0.000 (Baseline)	
<i>Post-Reform Horizon</i>						
2016	−0.671	0.508	−0.671	0.520	−0.671	0.508
2017	−2.059*	0.054	−2.059*	0.063	−2.059*	0.053
2018	−2.390**	0.028	−2.390**	0.034	−2.390**	0.027
2019	−0.994	0.331	−0.994	0.346	−0.994	0.331
2021	−1.690	0.108		—	−1.690	0.106
2022	−0.433	0.668	−0.433	0.677	−0.433	0.669
2023	−2.312**	0.033	−2.312**	0.039	−2.312**	0.032

Notes: Dynamic event-study coefficients ($\hat{\beta}_t$) relative to 2015. The reported p -values and significance stars are based on conventional OLS inference, uncorrected for the four-cluster structure, and describe the dynamic path rather than formal tests; formal inference relies on the Webb wild cluster bootstrap DiD estimates in Table 4.2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Source:* Author’s calculations based on MTA CIT registry data.

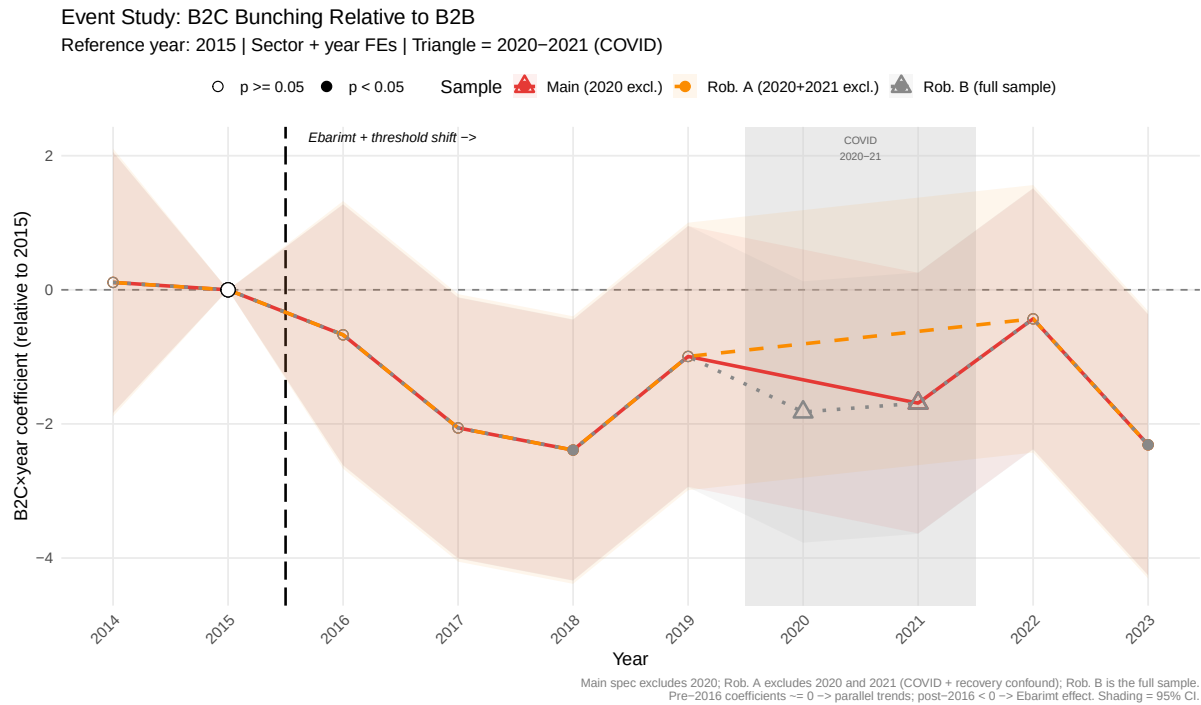
The effect builds gradually rather than appearing at the reform date. The 2016 coefficient is small (−0.671); divergence emerges in 2017 (−2.059) and grows through 2018 (−2.390). The path is not perfectly monotone, with the 2019 and 2022 coefficients pulling back toward zero, but it does not fade: the differential peaks in 2018 and the most recent estimate, $\hat{\beta}_{2023} = -2.312$, is essentially as large. A one-time 2016 compliance shock is hard to reconcile with this shape, while a progressively adopted consumer platform produces it naturally, a reading developed in Chapter 5. The coefficient p -values in Table 4.3 are conventional OLS, uncorrected for the four-cluster structure, so I read them as describing the dynamic path and the parallel-trends check rather than as formal tests. The wild cluster bootstrap delivers a valid p -value for the single Post×Treated coefficient, the one hypothesis the design is built to test; applying it year by year with four clusters would yield only coarse, low-powered values, so formal inference is reserved for the pooled DiD in Table 4.2.

The post-reform path runs through the pandemic years, and the three samples of Section 3.3 isolate how much the result depends on them. The 2020 estimates are noisy rather than strategic: retail $\hat{b}_{2020} = 1.581$ and food service 1.301 both fall below their 2019 values (2.220 and 2.565; Table A.4). The 2020 event-study coefficient appears only

in the full sample (-1.822 , $p = 0.083$) and is dropped from the Main and Robustness A specifications.

Manufacturing in 2021 is the key diagnostic. As a pure B2B sector with no ebarimt exposure, its bunching should follow macro conditions and threshold incentives, not consumer monitoring. Yet $\hat{b}_{2021} = 0.959$ is a 54% drop from the 2019 level of 2.096, with recovery to 2.063 by 2022. A V-shaped trough in an untreated sector is the signature of an external disruption, so the event-study coefficient $\hat{\beta}_{2021} = -1.690$ ($p = 0.108$) is best read as a noisy transitional-year signal. The Main specification keeps 2021; Robustness A drops it.

Figure 4.4: Event Study: B2C Differential Bunching Trajectories Across Alternative Sample Definitions



Notes: Event-study coefficients ($\hat{\beta}_t$) with 95% confidence intervals. Filled markers denote conventional OLS significance at the 5% level, open markers its absence; these are uncorrected for the four-cluster structure and describe the dynamic path, not formal tests (Table 4.2 carries the Webb wild cluster bootstrap inference). Triangles denote the COVID-19 years (2020–2021). The dashed line marks the January 2016 reform. *Source:* Author's calculations based on MTA CIT registry data.

Across the three samples the DiD coefficient is $\hat{\beta}_{\text{Main}} = -1.562$ ($N = 36$), $\hat{\beta}_{\text{Rob. A}} = -1.532$ ($N = 32$), and $\hat{\beta}_{\text{Full}} = -1.602$ ($N = 40$), a range of 0.070 units, under 5% around the central estimate. The Full sample is marginally the most negative and Robustness A the least; dropping 2021 removes a transitional year in which COVID disruption depressed manufacturing $\hat{b}$, and the resulting movement is small. The post-reform enforcement gap stays negative in every sample (-0.121 Main, -0.090 Robustness A, -0.160 Full), so the

sign inversion holds throughout. The Robustness A event study (Figure 4.4, dashed line) is nearly identical to the Main specification for all non-COVID years, with 2018 and 2023 again the coefficients furthest from zero ($\hat{\beta}_{2023} = -2.312$).

Control-group contamination. Nyamdavaa (2023) documents that ebarimt compliance gains among downstream B2C retailers spill upstream into wholesale reporting. Such partial treatment of the control depresses the B2B trajectory and biases $\hat{\beta}$ toward zero; consistent with this, wholesale $\hat{b}$ drifts up (2.033 in 2015 to 3.006 in 2023; Table A.3) rather than staying flat. This is the partial violation of Prediction 2 anticipated in Section 3.2: the B2B group is a clean counterfactual for the threshold shift only to the extent that it escapes consumer monitoring, and wholesale does not fully escape it. Manufacturing, with no direct link to B2C retail chains, is the cleaner control: restricting the control group to manufacturing alone yields $\hat{\beta} = -1.457$ (HC2 SE = 0.563, $p < 0.05$, $N = 27$; Appendix Table B.1). The slightly smaller magnitude arises because manufacturing itself declined modestly under the new 50M MNT notch, leaving the four-sector estimate the more conservative of the two.

Across all three sample definitions and both control-group choices, B2C bunching fell substantially relative to B2B after 2016, compressing the pre-reform enforcement gap from +1.441 to -0.121 (Table 4.2, column 1) and turning the sectoral differential negative by 2023 (Table 4.1). The primary estimate $\hat{\beta} = -1.562$ (Webb $p = 0.066$) is stable across specifications, and the dynamic pattern rather than a single significance star carries the identification.

Chapter 5

Discussion and Limitation

The Mongolian evidence suggests that the B2C enforcement gap is not a fixed feature of the credit-invoice VAT. Ebarimt works through a decentralized channel: consumer rebates and lottery entries turn ordinary retail sales into self-reported audit points, so verification scales with the spread of a consumer technology rather than with the size of the audit bureaucracy. The difference-in-differences results bear this out, with the pre-reform B2C–B2B gap closing and then inverting, and the event study shows the effect building over several years rather than at the reform date, which fits a process of gradual adoption (Section 4.3).

The sector pattern points the same way. The response was strongest where transactions are visible and technological frictions low: cash-heavy, face-to-face food service adjusted fastest, with retail following more gradually. One reform in one economy cannot establish how far this generalizes, but the direction is informative. Where sales are visible and the monitoring technology is easy to use, consumer-driven verification may be a low-cost complement to conventional audit capacity.

5.1 Implications for Tax Revenue

The design identifies how ebarimt reshaped the bunching distribution, not how much revenue it raised, and the two questions are worth keeping separate. A normalised bunching ratio measures the distortion around the threshold in standardised terms; it does not translate mechanically into revenue. For the revenue question the natural anchor is firm-level evidence from the same reform. Nyamdavaa (2023), using administrative VAT returns, finds that monitored retailers raised reported VAT liabilities by about 31 percent after ebarimt, with reported sales rising 42 percent, and that the effect spilled upstream, lifting suppliers' VAT liabilities by at least 15 percent. Both figures are lower bounds, since the wholesale control is itself partly treated. Those estimates, drawn from firm-level liabilities, are the appropriate basis for a revenue claim; the contribution here is to

show that the same reform compressed threshold bunching along the consumer-exposure margin the theory predicts.

The bunching estimates can still be given a rough revenue reading, useful for orders of magnitude even though it should not be treated as a precise figure. Appendix C sets out the calculation. Comparing post-reform B2C bunching with the counterfactual implied by the control-group trend, ebarimt appears to have moved on the order of one hundred consumer-facing firms per year out of the bunching region and toward compliance at the 50 million MNT threshold. A firm that registers rather than suppressing revenue remits output VAT of at least $\tau z^* = 5$ million MNT on the threshold base, and more once revenue above the threshold is counted, which places the additional VAT recovered from suppressed bunching on the order of 500 to 700 million MNT a year. Set against national VAT collections, which run into the trillions of MNT annually (National Statistics Office of Mongolia), that is a small fraction of one percent.

The contrast is the point rather than a shortfall. The bunching channel captures only firms clustered at the registration margin, a thin slice of the tax base, whereas the compliance gains Nyamdavaa (2023) documents operate across the full distribution of monitored firms and along the supply chain. The threshold response is best understood as a visible marker that consumer monitoring changed firm behaviour, not as a measure of the reform’s total revenue yield. A full welfare account would also net the programme’s costs, the consumer rebate of one-fifth of the VAT paid, the lottery prizes, and the point-of-sale and data infrastructure, against revenue gains of the kind Nyamdavaa estimates; that accounting lies beyond this paper and is a natural use of firm-level data. The response is also larger and cleaner for the VAT than for taxes firms can adjust on the cost side: Nyamdavaa (2023) finds that monitored retailers offset much of their higher reported sales by inflating costs on income tax returns, so income tax liabilities rose far less than VAT liabilities.

This bears directly on current policy. A standing parliamentary proposal would raise the registration threshold further. The evidence here counsels caution, since a higher cutoff would release from the VAT net precisely the small consumer-facing firms that ebarimt has made comparatively cheap to monitor, conceding the enforcement gain the reform achieved at that margin.

5.2 Limitations

Several constraints bound these conclusions. The data are binned and aggregated, so they cannot track individual firms or separate the bunching response into real-output adjustment and strategic misreporting. Because the estimator works on sector aggregates rather than firms, and inference rests on only four sector clusters, the confidence intervals around the estimates are wide: the results establish the direction and the rough magnitude

of the response rather than a tightly bounded coefficient. For the same reason, treatment is identified at the sector aggregate: the estimates show how the bunching distribution shifted within B2C relative to B2B, not how firms differed within each group. The outcome itself is a normalized bunching ratio, which measures how far the distortion moved in standardized terms rather than the share of firms that changed behaviour, and the data do not allow conversion between the two.

Identification is limited in three ways. The threshold increase and the ebarimt launch took effect together in 2016 and are best treated as a single regime change; the design does not separate them by timing, and since the higher cutoff applied to both groups, only the residual cross-sector difference is attributed to consumer monitoring. Section 4.3 argues that the threshold move was a sector-neutral adjustment absorbed by the year fixed effects, and that the compositional channel by which it could still bias the estimate is not visible in the threshold-zone data; but the common-threshold assumption, that the fivefold increase moved B2C and B2B bunching by the same amount, remains an assumption rather than a separately tested fact, because the binned data cannot compare the two groups' full revenue distributions near the notch. The parallel-trends test is also limited, because only one pre-reform coefficient is estimable; that coefficient, $\hat{\beta}_{2014}$, is indistinguishable from zero (Section 4.3), which is consistent with parallel trends but far from proof. And the control group is exposed to cross-sector spillovers: if ebarimt raised B2B supplier compliance through invoice-matching pressure (Nyamdavaa 2023), the control is partially treated and $\hat{\beta}$ understates the true effect. Restricting the control to manufacturing, which has no direct retail supply link, leaves the estimate negative and of similar magnitude (Section 4.3).

Inference is constrained in two further ways. With four sector clusters, conventional cluster-robust standard errors over-reject, so I use the wild cluster bootstrap with Webb six-point weights; the resulting p -values are small-sample approximations, and the case rests on the stability of the point estimate across specifications as much as on significance levels. The bunching ratios also enter the difference-in-differences as estimated rather than observed outcomes. The within-cell bootstrap captures their first-stage uncertainty, but a joint two-step treatment in the spirit of Arkhangelsky, Yanagimoto, and Zohar (2024) is infeasible with four groups, so the reported standard errors are conditional on the first-stage estimates. Those first-stage errors are tight for the control sectors and for retail throughout, while the food service cells in the early post-reform years caution against over-reading any single coefficient.

These limits indicate the natural next steps, and they are also what would raise the study to the standard of a journal contribution. Firm-level panel data would follow individual firms across the threshold, separating real output responses from strategic misreporting, measuring directly how many firms adjusted, and delivering the narrower confidence intervals that the sector-level aggregates cannot. Regional variation in ebarimt's

rollout and enforcement intensity would supply a second source of identification and test whether the compliance response scales with local enforcement. With data of that kind the present sector-level design could be sharpened to the firm or region level. The contribution here should be read accordingly: not a final accounting of ebarimt's effect, but an identification of it within real data constraints, and a basis for the more granular work that richer data will allow.

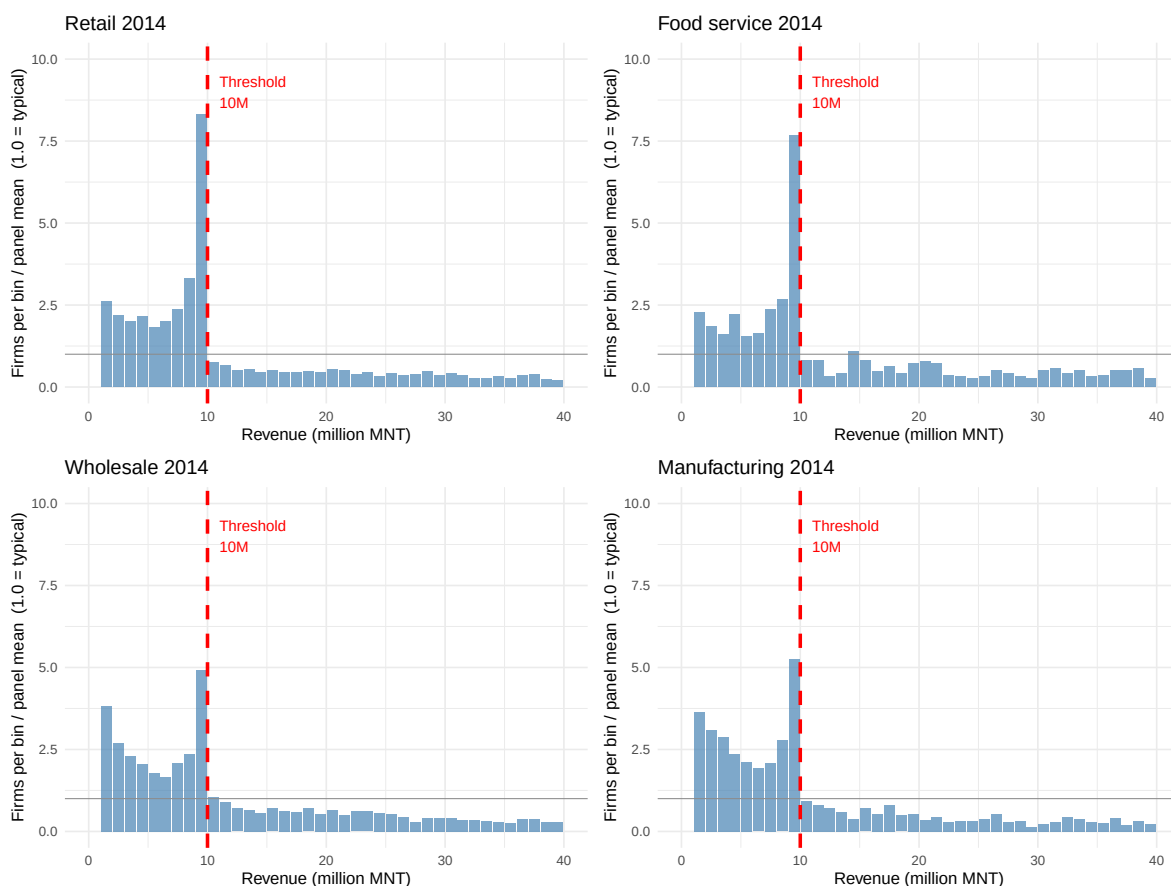
Appendix

A Baseline Parameter Profiles and COVID-Year Estimations

This appendix collects the year-by-year estimates behind the main results: the raw sector-year distributions, the 2020 and 2021 (COVID) estimates, and the full 36-cell table of $\hat{b}_{st}$.

Figure A.1: Revenue Distributions at the VAT Threshold: 2014 (Pre-Reform)

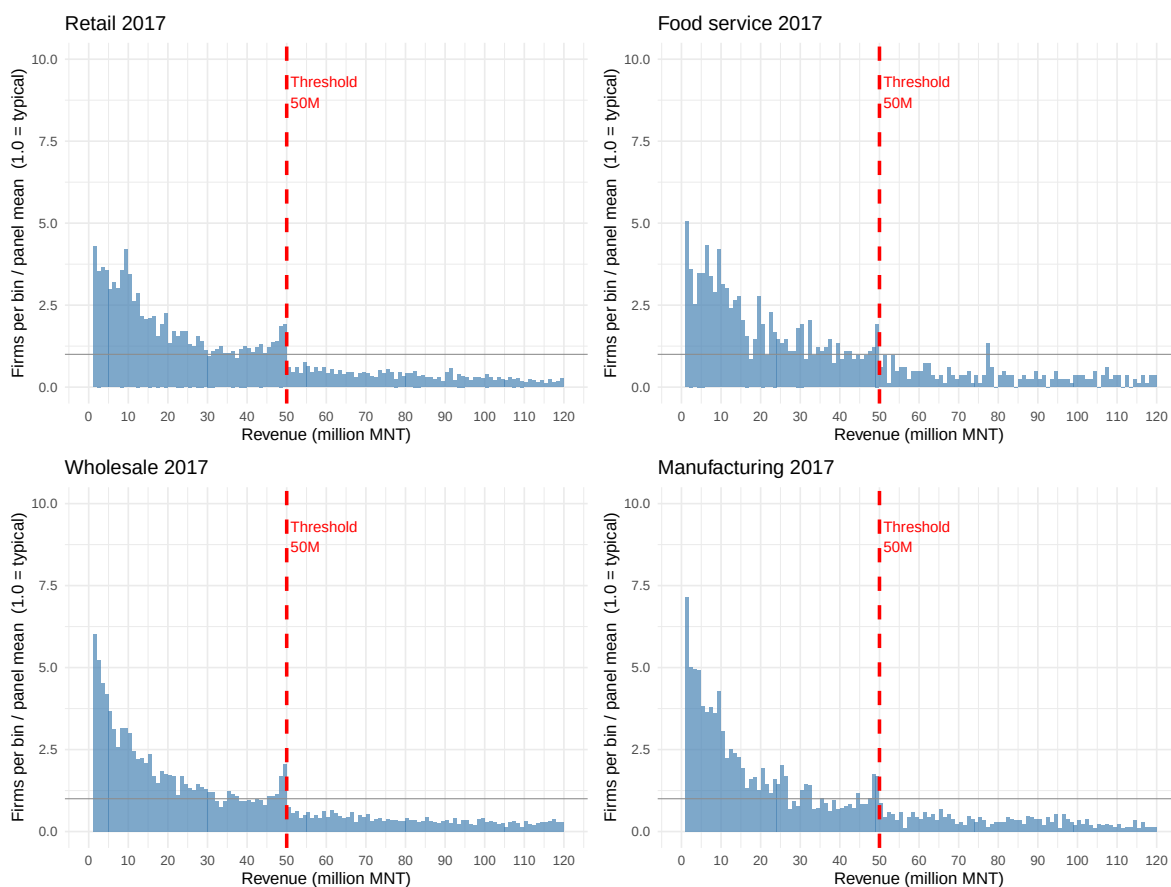
Raw revenue distributions -- 2014
Red dashed line = VAT registration threshold



Source: MTA CIT registry data.

Figure A.2: Revenue Distributions at the VAT Threshold: 2017 (Post-Reform)

Raw revenue distributions -- 2017
Red dashed line = VAT registration threshold



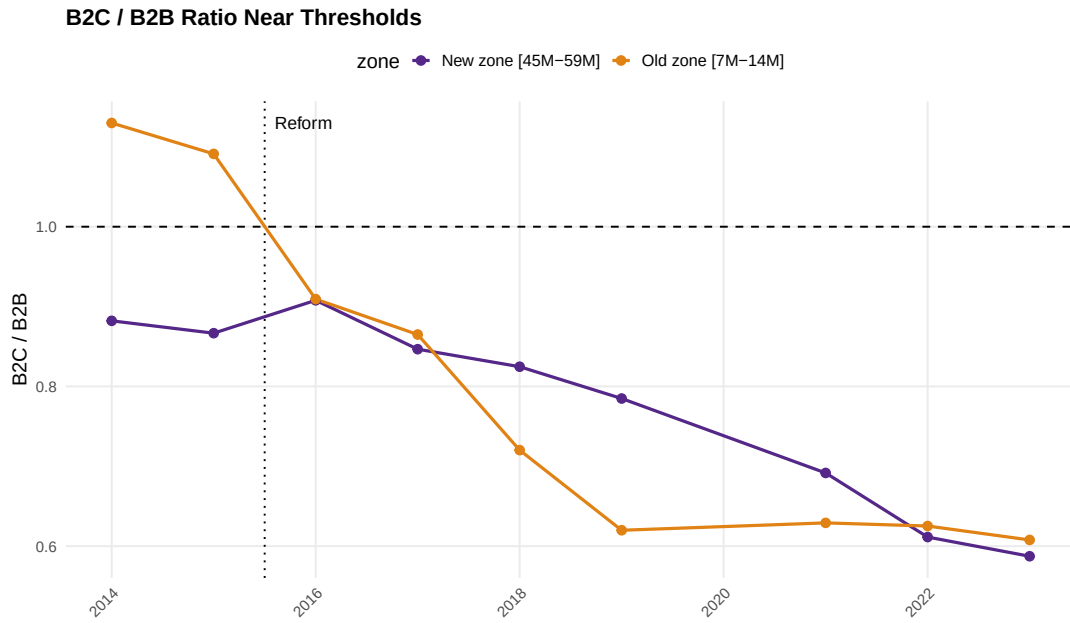
Notes: Raw firm counts by 1M MNT revenue bins for each sector. The red dashed line is a VAT registration threshold at 50M MNT. The catch-all bin [0, 1,000,000) MNT is excluded. *Source:* MTA CIT registry data.

Table A.1: **Revenue Density Discontinuities at the Active Registration Threshold**

	<i>Pre-Reform: $z_1^* = 10M$ MNT (2014)</i>			<i>Post-Reform: $z_2^* = 50M$ MNT (2017)</i>		
Sector	N_{j-} ^a	N_{j+} ^b	Ratio ^c	N_{j-} ^a	N_{j+} ^b	Ratio ^c
<i>B2C Treatment Group</i>						
Retail Trade	1,105	98	11.3	101	31	3.3
Food Services	149	16	9.3	16	5	3.2
<i>B2B Control Group</i>						
Wholesale Trade	582	123	4.7	118	42	2.8
Manufacturing	271	48	5.6	35	18	1.9

Source: MTA CIT registry data. ^a Firm count in the 1M MNT bin immediately below the threshold ($[z_t^* - 1M, z_t^*)$). ^b Firm count in the 1M MNT bin immediately above the threshold ($[z_t^*, z_t^* + 1M)$). ^c Density ratio N_{j-}/N_{j+} ; values near 1 indicate no discontinuity.

Figure A.3: **B2C/B2B Firm Composition Near VAT Thresholds, 2014–2023**



Source: MTA CIT registry data.

Table A.2: **Threshold-Zone Firm Counts and B2C/B2B Ratios by Year**

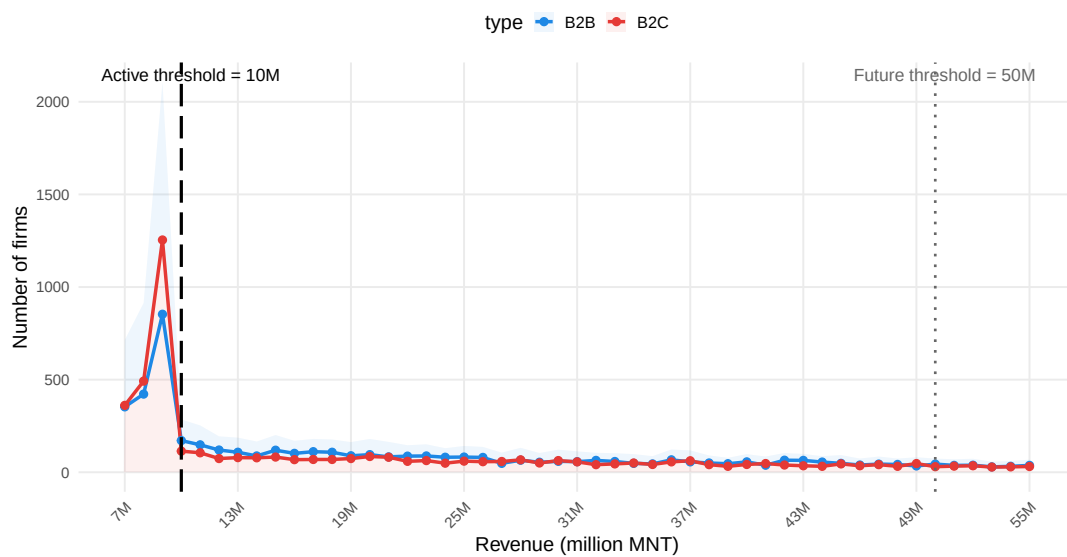
Year	<i>Old zone [7M, 15M)</i>			<i>New zone [45M, 60M)</i>		
	B2C	B2B	Ratio	B2C	B2B	Ratio
2014	2,557	2,263	1.13	494	560	0.88
2015	2,261	2,072	1.09	507	585	0.87
2016	1,605	1,765	0.91	729	803	0.91
2017	1,461	1,689	0.87	795	939	0.85
2018	1,228	1,705	0.72	875	1,061	0.82
2019	1,073	1,731	0.62	923	1,176	0.78
2020	981	1,560	0.63	821	1,149	0.71
2021	860	1,367	0.63	812	1,174	0.69
2022	869	1,390	0.63	829	1,356	0.61
2023	846	1,392	0.61	918	1,563	0.59

Source: MTA CIT registry data. Counts are total active firms with lower bin edge in the stated revenue zone, summed across the two B2C sectors (retail, food service) and the two B2B sectors (wholesale, manufacturing). Ratio is B2C/B2B.

Figure A.4: Pre-Reform Firm Distributions

2014 (Pre-Reform): Full Distribution

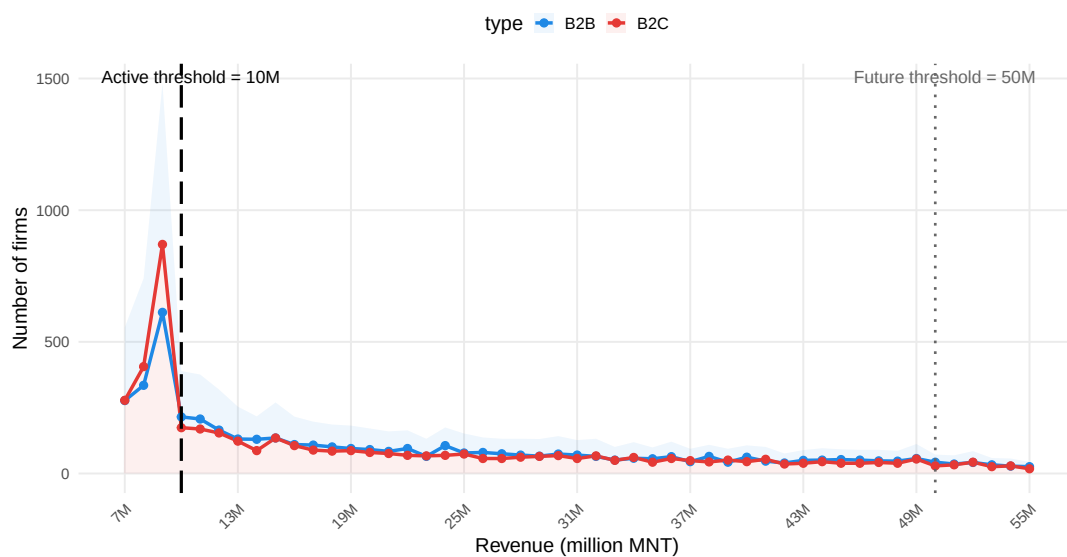
No bunching at 50M before reform



(a) 2014: Full distribution (7M–55M)

2015 (Pre-Reform): Full Distribution

No bunching at 50M before reform

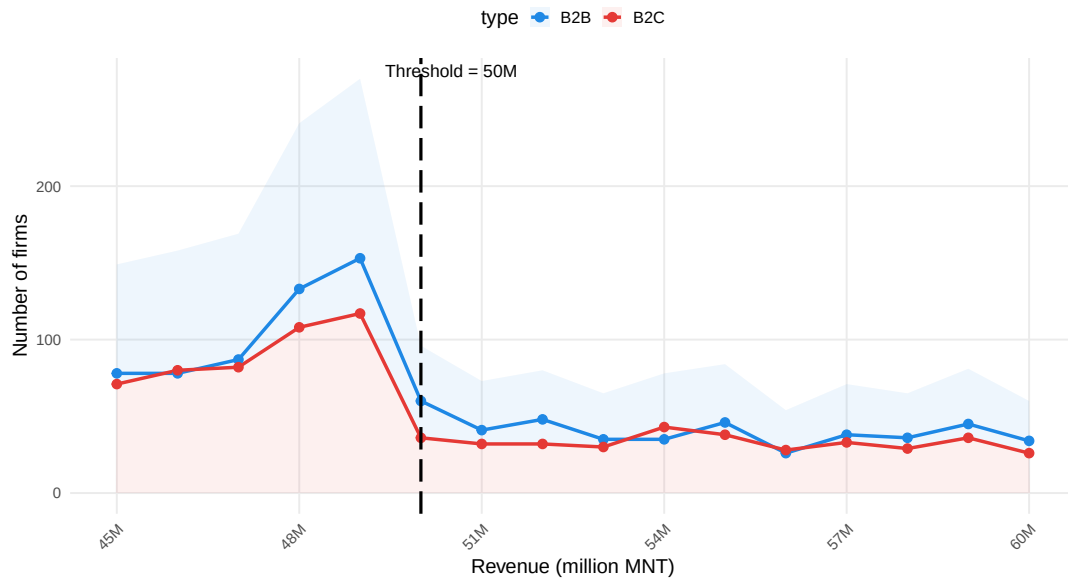


(b) 2015: Full distribution (7M–55M)

Source: MTA CIT registry data.

Figure A.5: Post-Reform Firm Distributions

2017: Distribution Around 50M Threshold



(a) 2017: Around 50M threshold

2022: Distribution Around 50M Threshold



(b) 2022: Around 50M threshold

Source: MTA CIT registry data.

Table A.3: **Normalised Bunching Estimates $\hat{b}_{st}$ by Sector and Year (Main Sample)**

Sector	2014	2015	2016	2017	2018	2019	2021	2022	2023
<i>B2C Treatment Group</i>									
Retail	4.368† (0.263)	4.050† (0.189)	2.089 (0.169)	1.755 (0.167)	2.417 (0.210)	2.220 (0.183)	1.487 (0.154)	1.919 (0.167)	1.703 (0.150)
Food Service	3.945† (0.430)	2.411† (0.310)	2.282 (0.575)	1.579 (0.415)	1.257 (0.284)	2.565 (0.469)	0.564 (0.409)	4.099 (1.633)	1.585 (0.455)
<i>B2B Control Group</i>									
Wholesale	2.301 (0.150)	2.033† (0.121)	1.989 (0.177)	2.088 (0.160)	2.017 (0.183)	1.904 (0.141)	1.700 (0.159)	2.048 (0.181)	3.006 (0.178)
Manufacturing	3.018† (0.215)	1.655† (0.148)	0.952 (0.259)	2.592† (0.462)	3.665† (1.009)	2.096 (0.284)	0.959 (0.226)	2.063 (0.241)	2.133 (0.273)
<i>B2C Mean</i>	4.156	3.230	2.185	1.667	1.837	2.392	1.026	3.009	1.644
<i>B2B Mean</i>	2.660	1.844	1.471	2.340	2.841	2.000	1.330	2.056	2.570
<i>Gap (B2C – B2B)</i>	+1.497	+1.386	+0.715	–0.673	–1.004	+0.393	–0.304	+0.953	–0.926

Notes: $\hat{b}_{st}$ from a 7-degree polynomial counterfactual. Exclusion window $[z_t^* - 2M, z_t^* + 30M)$ post-reform, $[z_t^* - 2M, z_t^* + 10M)$ pre-reform. Bootstrap standard errors (500 replications) in parentheses. Negative values indicate deficits relative to the counterfactual. 2020 excluded from this table.

† Missing mass did not reach bunching mass within the cap Capped cells. *Source:* Author’s calculations based on MTA CIT registry data.

Table A.4: **Bunching Estimates in Pandemic Years (2020 and 2021)**

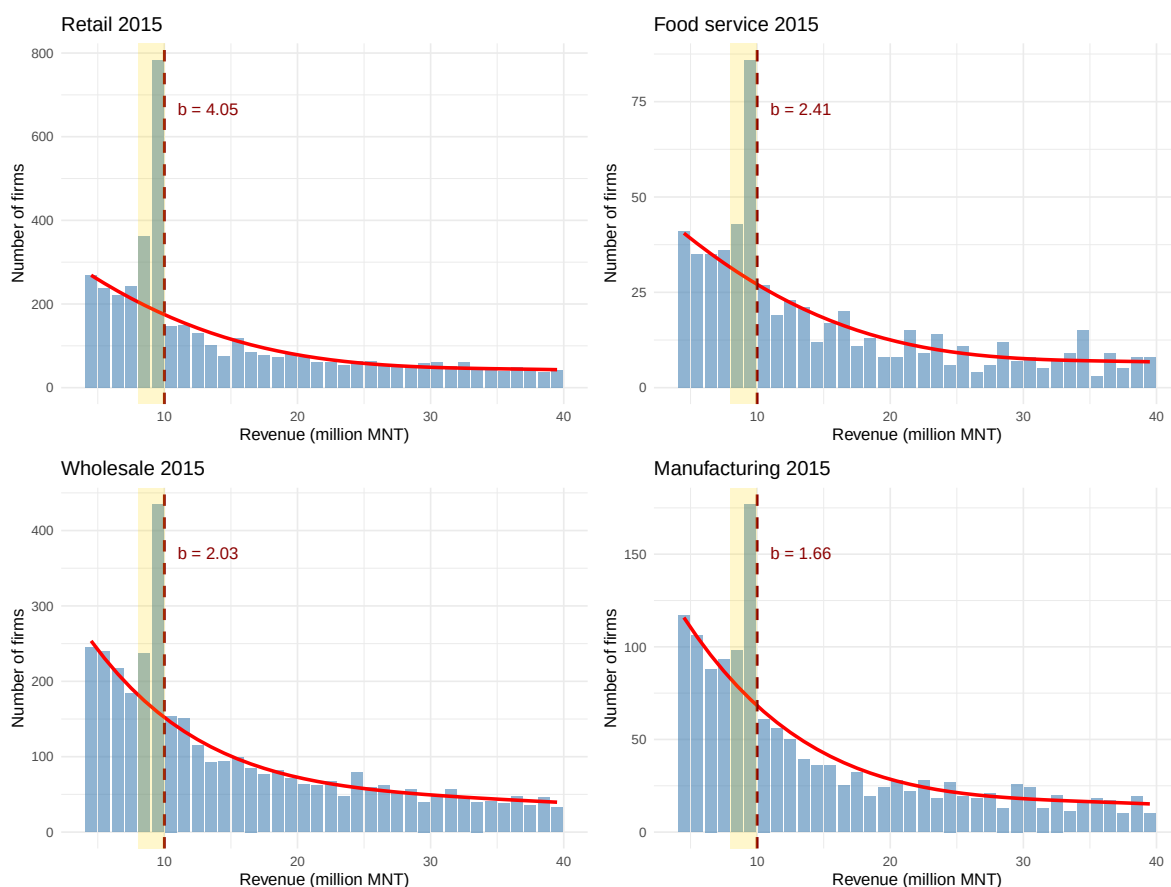
Sector	2020			2021		
	$\hat{b}$	SE	95% CI	$\hat{b}$	SE	95% CI
Retail (B2C)	1.581	0.181	[1.235, 1.957]	1.487	0.154	[1.197, 1.790]
Food Service (B2C)	1.301	0.382	[0.693, 2.222]	0.564	0.409	[-0.078, 1.490]
Wholesale (B2B)	1.119	0.147	[0.855, 1.418]	1.700	0.159	[1.420, 2.047]
Manufacturing (B2B)	2.635	0.635	[1.705, 4.081]	0.959	0.226	[0.593, 1.500]
<i>B2C mean</i>	1.441			1.026		
<i>B2B mean</i>	1.877			1.330		
<i>Gap (B2C – B2B)</i>	–0.436			–0.304		

Notes: 2020 observations are excluded from the Main and Robustness A samples but retained in the Full sample. 2021 observations are included in Main but excluded from Robustness A. Bootstrap standard errors based on 500 replications. *Source:* Author’s calculations based on MTA CIT registry data.

Figure A.6: Observed Distribution and Polynomial Counterfactual: 2015 (Pre-Reform)

Actual vs Counterfactual -- 2015

Red = counterfactual | Gold = bunching window | b = excess mass

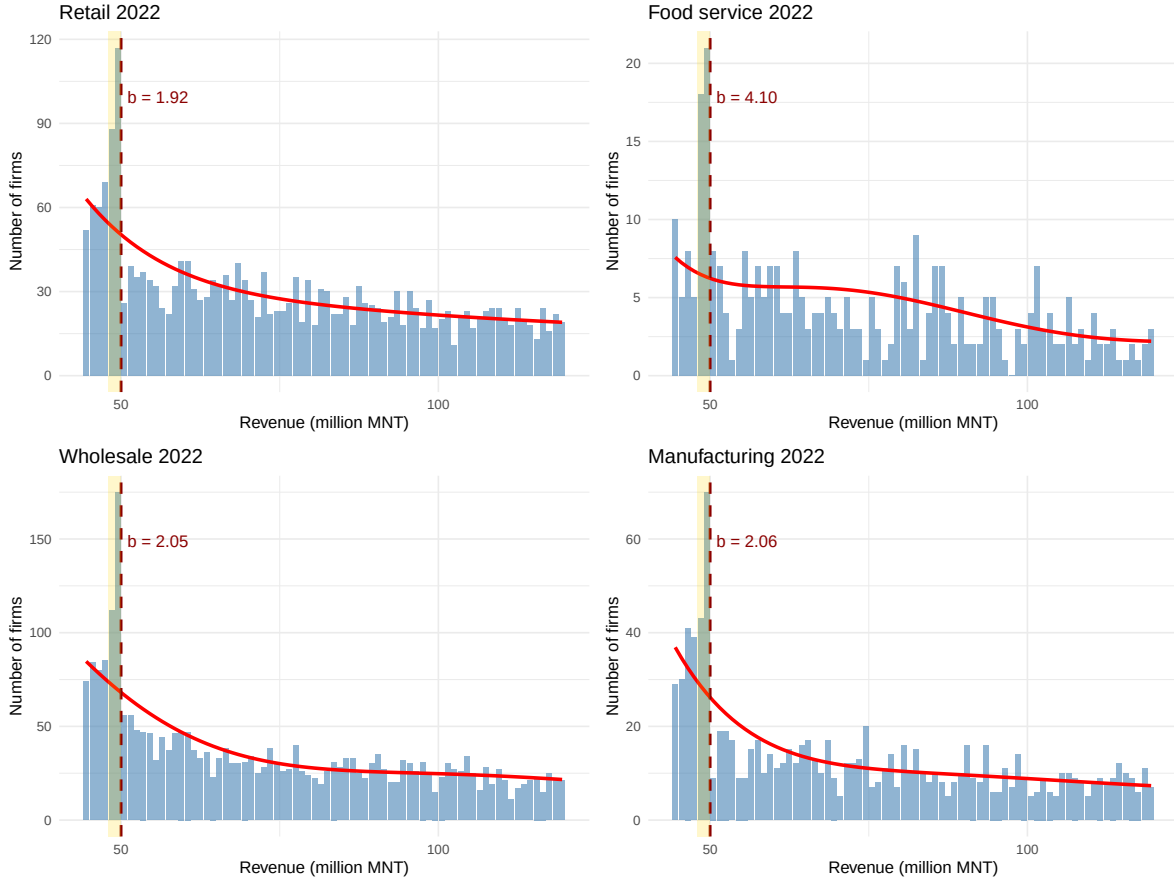


Notes: Observed firm counts by revenue bin (bars) are plotted alongside the fitted polynomial counterfactual (red line). The shaded region denotes the exclusion window (R_t) around the VAT registration threshold. The active threshold is $z^* = 10\text{M MNT}$. Excess mass below the threshold relative to the counterfactual identifies bunching behavior. *Source:* Author's calculations based on MTA CIT registry data.

Figure A.7: Observed Distribution and Polynomial Counterfactual: 2022 (Post-Reform)

Actual vs Counterfactual -- 2022

Red = counterfactual | Gold = bunching window | b = excess mass



Notes: Observed firm counts by revenue bin (bars) are plotted alongside the fitted polynomial counterfactual (red line). The shaded region denotes the exclusion window (R_t) around the VAT registration threshold. The active threshold is $z^* = 50$ M MNT. Relative to 2015, reduced excess mass below the threshold is consistent with improved enforcement following the introduction of ebarimt.

Source: Author's calculations based on MTA CIT registry data.

B Robustness and Sensitivity Checks

This appendix reports the robustness checks referenced in Chapter 4: the DiD re-estimated with manufacturing as the sole control, and the sensitivity of the bunching estimates to the polynomial degree (Table B.2) and the bunching-window width (Table B.3).

Manufacturing-only control group. Table B.1 re-estimates the DiD with manufacturing as the only control. Dropping wholesale removes the control sector most exposed to upstream spillovers (Section 4.3), leaving a comparison group with no consumer-facing sales. The estimate stays negative and close to the four-sector benchmark across all three samples ($\hat{\beta}$ between -1.457 and -1.593 , against -1.562 in the main specification).

Table B.1: **Difference-in-Differences Sensitivities: Manufacturing-Only Control Group**

	(1) Main (2020 excl.)	(2) Robustness A (2020+21 excl.)	(3) Full Sample (all years)
$\hat{\beta}$: Post $\times$ Treated	-1.457^{**} (0.563)	-1.484^{**} (0.627)	-1.593^{***} (0.538)
Sector + Year FE	Yes	Yes	Yes
N (cells)	27	24	30

Notes: HC2 heteroskedasticity-robust standard errors in parentheses (wild cluster bootstrap not applied here as $G = 3$); with three clusters HC2 inference is anti-conservative, so the markers are indicative and formal inference rests on the four-cluster bootstrap in Table 4.2. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Treatment group: B2C (retail, food services). Control group: manufacturing only (wholesale excluded). *Source:* Author's calculations based on MTA CIT registry data.

Sensitivity to polynomial degree. Table B.2 varies the counterfactual polynomial from degree 5 to degree 8, holding all else fixed. The estimates are stable across degrees 5 to 7 for nearly every cell. The 2022 food service reading ($\hat{b} = 4.099$) is an outlier against the sector's declining trend, resting on a sparse near-threshold distribution: it holds across degrees 5 to 7 but overfits to 6.532 at degree 8, which marks the cell as fragile (Appendix Table B.2). The baseline degree-7 specification balances flexibility against this risk.

Table B.2: **Sensitivity of Bunching Estimates to Polynomial Degree: Selected Sector-Year Cells**

Sector-Year	Degree 5	Degree 6	Degree 7 (Baseline, $p = 7$)	Degree 8
<i>Panel A: Pre-Reform (2015, $z^* = 10M$ MNT)</i>				
Retail 2015	4.050	3.790	3.820	3.652
Food service 2015	2.411	2.180	2.161	1.994
Wholesale 2015	2.033	2.069	2.120	2.341
Manufacturing 2015	1.655	1.665	1.676	1.575
<i>Panel B: Post-Reform (2022, $z^* = 50M$ MNT)</i>				
Retail 2022	1.881	1.890	1.919	1.839
Food service 2022	3.678	3.618	4.099	6.532

Notes: Exclusion window $[z_t^* - 2M, z_t^* + 30M)$ post-reform, $[z_t^* - 2M, z_t^* + 10M)$ pre-reform. Degree 7 is the baseline specification. Source: Author's calculations based on MTA CIT registry data.

Sensitivity to window width. Table B.3 varies the bunching window below the threshold. The baseline 2-bin window ($[z_t^* - 2M, z_t^*)$) gives $\hat{\beta} = -1.562$; widening to 3 bins keeps the negative sign ($\hat{\beta} = -0.974$) and an 80% post-reform narrowing. At 4 bins and beyond the window reaches into the rising pre-notch slope of the small-firm density, the pre-reform gap turns sharply negative, and the estimates break down. A window of $\delta \in \{2, 3\}$ is therefore the admissible range, as Section 3.4 notes.

Table B.3: **Sensitivity of DiD Estimate to Bunching Window Width (excl_lo Sweep)**

δ^a	B2C pre	B2C post	B2B pre	B2B post	Gap pre	Gap post	DiD	% narrow
2 [†]	3.693	1.966	2.252	2.087	+1.441	-0.121	-1.562	108%
3	4.703	2.542	3.481	2.294	+1.222	+0.248	-0.974	80%
4	17.078	2.569	97.370	2.512	-80.291	+0.056	80.348	100%
5	26.216	2.856	105.689	3.203	-79.473	-0.347	79.126	100%

Notes: Each row re-estimates all sector-year cells with a bunching window of $[z_t^* - \delta M, z_t^*)$. DiD = $(\Delta \bar{b}_{B2C}) - (\Delta \bar{b}_{B2B})$, main sample (2020 excluded). % narrowing = $(\text{gap}_{\text{pre}} - \text{gap}_{\text{post}}) / \text{gap}_{\text{pre}} \times 100$. [†] Locked specification ($\delta = 2$). ^a Window width in 1M MNT bins below the threshold. DiD sign is stable at $\delta \in \{2, 3\}$; larger windows extend into the ascending pre-notch slope and are not reported as admissible. Source: Author's calculations based on MTA CIT registry data.

C An Illustrative Revenue Calculation

This appendix details the back-of-envelope revenue figure discussed in Section 5.1. It is illustrative and rests on assumptions the binned data cannot test; the firm-level estimates of Nyamdavaa (2023) remain the appropriate basis for revenue claims.

The calculation has two inputs: the number of consumer-facing firms ebarimt moved out of bunching each year, and the VAT each such firm remits once compliant.

For the first, I compare, in each post-reform year, the observed B2C excess mass with the counterfactual that would have prevailed had the pre-reform enforcement gap persisted. Taking the contemporaneous B2B bunching intensity and adding the pre-reform B2C–B2B gap of 1.441 standardised units gives the counterfactual B2C intensity absent ebarimt; the difference from the observed intensity, converted to a firm count using the fitted counterfactual density in the bunching window, averages roughly one hundred B2C firms per year moved toward compliance over 2016–2023 (2020 excluded).

For the second, a firm that registers at the 50 million MNT threshold rather than suppressing revenue below it remits output VAT of at least $\tau z^* = 0.10 \times 50 = 5$ million MNT on the threshold base, and more once revenue above the threshold is counted; the missing-mass region above the notch places representative compliant revenue near 65 million MNT, or about 6.5 million MNT in VAT. Multiplying the annual firm count by these per-firm amounts yields additional VAT of approximately 500 to 700 million MNT per year.

Two caveats bound the figure. It assumes de-bunched firms comply fully and that suppressed revenue reflects evasion rather than real output restraint, neither of which the aggregated data can verify. And it counts only the registration margin, omitting the intensive-margin compliance gains across the wider firm distribution that firm-level data capture. Against national VAT revenue in the trillions of MNT (National Statistics Office of Mongolia), the recovered amount is on the order of a few hundredths of one percent, a reminder that the bunching response identifies a mechanism rather than a large aggregate revenue effect.

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